

Brain-wide arousal signals are segregated from movement planning in the superior colliculus

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This study presents a **valuable** finding relating to how the state of arousal is represented within the superior colliculus, a principal visuo-oculomotor structure. The main conclusion that the representation of arousal is segregated, and thus influences visual activity but not motor output, is **incompletely** supported by the evidence, but could be stronger if a specific concern relating to an alternative explanation for the dichotomy was addressed. The work will be of interest to sensory, motor, and cognitive neuroscientists.

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Abstract

The superior colliculus (SC) is traditionally considered a brain region that functions as an interface between processing visual inputs and generating eye movement outputs. Although its role as a primary reflex center is thought to be conserved across vertebrate species, evidence suggests that the SC has evolved to support higher-order cognitive functions including spatial attention. When it comes to oculomotor areas such as the SC, it is critical that high precision fixation and eye movements are maintained even in the presence of signals related to ongoing changes in cognition and brain state, both of which have the potential to interfere with eye position encoding and movement generation. In this study, we recorded spiking responses of neuronal populations in the SC while monkeys performed a memory-guided saccade task and found that the activity of some of the neurons fluctuated over tens of minutes. By leveraging the statistical power afforded by high-dimensional neuronal recordings, we were able to identify a low-dimensional pattern of activity that was correlated with the subjects' arousal levels. Importantly, we found that the spiking responses of deep-layer SC neurons were less correlated with this brain-wide arousal signal, and that neural activity associated with changes in pupil size and saccade tuning did not overlap in population activity space with movement initiation signals. Taken together, these findings provide a framework for understanding how signals related to cognition and arousal can be embedded in the population activity of oculomotor structures without compromising the fidelity of the motor output.

Introduction

For humans and animals to effectively interact with a fast-changing environment, they must be adept at performing a range of visually-guided motor actions. Basic everyday tasks that are critical for our survival (e.g. making an eye movement to a car that is about to cross our path) require visual information to be rapidly encoded and transformed into a motor action. In non-mammals such as birds, reptiles and fish, visually-guided motor actions (e.g. prey capture) are generated by circuits in the optic tectum (Wylie et al., 2009 [↗](#); Kardamakis et al., 2015 [↗](#); Suzuki et al., 2019 [↗](#); Allen et al., 2021 [↗](#); Isa et al., 2021 [↗](#)) whereas in mammals such as rodents, cats and monkeys they are supported by activity in a homologous structure called the superior colliculus (SC) (Wurtz and Albano, 1980 [↗](#); Sparks and Mays, 1990 [↗](#); Gandhi and Katnani, 2011 [↗](#); Cang et al., 2018 [↗](#); Wheatcroft et al., 2022 [↗](#); Hafed et al., 2023 [↗](#)).

The SC resides in the roof of the brainstem and receives input from retinal ganglion cells (Perry and Cowey, 1984 [↗](#); Qu et al., 2006 [↗](#)), the primary visual cortex, and extrastriate areas such as V4 and the middle temporal area (Cerkevich et al., 2014 [↗](#)). In addition to these visual inputs, research has shown that it also receives projections from cortical regions that have been implicated in eye movement control including the supplementary eye field (Huerta and Kaas, 1990 [↗](#); Parthasarathy et al., 1992 [↗](#)) and the frontal eye field (Komatsu and Suzuki, 1985 [↗](#); Stanton et al., 1988 [↗](#); Parthasarathy et al., 1992 [↗](#)). The role of these projections in the visuomotor transformation depends on the functional layer of the SC in which they terminate. Decades of research has shown that the SC comprises three main layers each of which contains cells with distinct neurophysiological response properties (May, 2006 [↗](#); Krauzlis et al., 2013 [↗](#); Basso and May, 2017 [↗](#)). For example, neurons in the superficial layers, which are thought to represent the earliest stage of the visuomotor transformation, only respond to visual stimuli presented within their receptive field (RF) (Goldberg and Wurtz, 1972 [↗](#)). In contrast, neurons in the intermediate layers elicit a response during both the presentation of visual stimuli and when saccades are made into their RF. Finally, neurons in the deep layers primarily respond in and around the time of a saccade. As is the case for neurons in the intermediate layers, they form a topographically organized map of visual space (Robinson, 1972 [↗](#); Katnani and Gandhi, 2011 [↗](#); White et al., 2017 [↗](#)) and encode saccades across a range of directions and amplitudes (Gandhi and Katnani, 2011 [↗](#)). Deep-layer SC neurons project to downstream nuclei innervating the extraocular muscles (e.g. the central mesencephalic reticular formation) (Sparks, 1986 [↗](#); Sparks and Hartwich-Young, 1989 [↗](#); Bohlen et al., 2017 [↗](#)) but thresholds for evoking saccades using electrical microstimulation are often lowest in the intermediate layers (Mohler and Wurtz, 1976 [↗](#)) suggesting that the intermediate and deep layers of the SC function together to control saccadic eye movements (Wurtz and Albano, 1980 [↗](#)).

Although the SC is best known for its role in visually-guided motor actions, our understanding of its function has evolved considerably and recent work has demonstrated its role in processes linked to higher order cognition (Knudsen, 2011 [↗](#); Krauzlis et al., 2013 [↗](#); Clark et al., 2015 [↗](#); Basso et al., 2021 [↗](#)). For example, behavioral performance on covert attention tasks (in which subjects are cued to attend to a stimulus presented at a specific portion of visual space) is modulated when the activity of SC neurons with RFs at the cued location are perturbed using reversible inactivation or microstimulation (Müller et al., 2005 [↗](#); Zénon and Krauzlis, 2012 [↗](#)). In addition to cognitive signals, the SC is interconnected with brain regions that control arousal levels, e.g. the locus coeruleus (LC) (Edwards et al., 1979 [↗](#); Aston-Jones and Cohen, 2005 [↗](#); Sara and Bouret, 2012 [↗](#); Li et al., 2018 [↗](#); van den Brink et al., 2019 [↗](#); Plummer et al., 2020 [↗](#); Benavidez et al., 2021 [↗](#)). Several studies using microstimulation have been able to establish a link between SC activity and non-invasive markers of arousal such as pupil size (Wang et al., 2012 [↗](#), 2014 [↗](#); Joshi et al., 2016 [↗](#); Wang and Munoz, 2021 [↗](#)). Furthermore, research has shown that oculomotor behavior is impacted by changes in arousal (Di Stasi et al., 2013 [↗](#)). For example,

participants are faster at initiating visually-guided saccades to peripheral target stimuli when a startling auditory stimulus is presented prior to the onset of the target (Castellote et al., 2007 [↗](#)). These changes, which are thought to reflect increased levels of arousal, are accompanied by increased pupil size and decreased microsaccade rate even when participants perform auditory tasks that do not necessitate saccadic eye movements (Contadini-Wright et al., 2023 [↗](#)).

To support signals related to higher-order cognition, the SC has had to undergo a number of adaptations (Basso et al., 2021 [↗](#)). The precise details of how conserved neuronal cell types give rise to complex processes such as spatial attention has yet to be fully elucidated. However, it must occur in a manner that does not interfere with the signals needed for volitional control of the eyes be it in the context of stable fixation or large amplitude saccades. As is the case for cortical regions that have a direct influence on motor behavior, having signals related to higher-order cognition and brain state intermingled with those used for movement encoding could be problematic as it may lead to interference with motor planning via premature or unwanted muscle contractions. In the motor cortex, research using delayed-reach tasks has shown that population responses during movement preparation are kept separate from those used during movement initiation (Kaufman et al., 2014 [↗](#); Elsayed et al., 2016 [↗](#)). Furthermore, visual and motor responses in the SC occupy separate subspaces when they are recorded during the same oculomotor task and across different tasks which could facilitate the decoding of signals related to sensation, action and context by downstream regions (Jagadisan and Gandhi, 2022 [↗](#); Ayar et al., 2023 [↗](#)). We hypothesized that a similar mechanism might exist in the SC to prevent arousal-related fluctuations from interfering with eye movement outputs.

In this study, we observed arousal-related fluctuations in the spiking responses of neuronal populations in the SC while monkeys performed a task that is commonly used to study visually-guided motor actions. By using dimensionality reduction on the neural population data, we were able to identify an axis in population activity space that was similar to the “slow drift axis” that has been observed in several cortical areas (Cowley et al., 2020 [↗](#)) and is known to be associated with a constellation of arousal-related variables including pupil size and alpha power as measured using scalp electroencephalography (EEG) (Johnston et al., 2022a [↗](#), 2022b [↗](#)). Importantly, we found evidence that this signal is isolated from SC neurons with strong saccadic responses: the activity of these cells was less correlated with the slow drift axis when it was computed using population activity in the SC and simultaneously recorded data in the prefrontal cortex (PFC). Finally, we were able to show that signals in the SC that are used to control the size of the pupil reside in an orthogonal subspace to those that are used for saccade generation. This provides a potential mechanism through which signals related to cognition and arousal can exist in the SC, and even impact processing at the early to mid-stages of the visuomotor transformation, without leading to unwanted changes in SC neurons that are most active around the time of a saccade.

Results

To determine how arousal-related signals related to motor-related activity in the SC, we trained two monkeys to perform a memory-guided saccade task (MGS) task (Figure 1A [↗](#)). During the task, subjects maintained fixation on a central point while a brief target stimulus was presented in the periphery. The subjects had to remember the location of this stimulus during a delay period and make a saccade to that location when the fixation point was extinguished. While the subjects performed the task (Monkey Do = 25 sessions, Monkey Wa = 26 sessions), we recorded the spiking responses of neuronal populations in: 1) the SC using a 16-channel linear microelectrode array; and 2) the PFC (area 8Ar) using a 96-channel “Utah” array (Figure 1B [↗](#)).

To determine the neurophysiological response properties of individual SC neurons, we took advantage of the fact that the MGS task contains distinct epochs that can dissociate visual responses from those occurring during the execution of saccadic eye movements (Hikosaka and

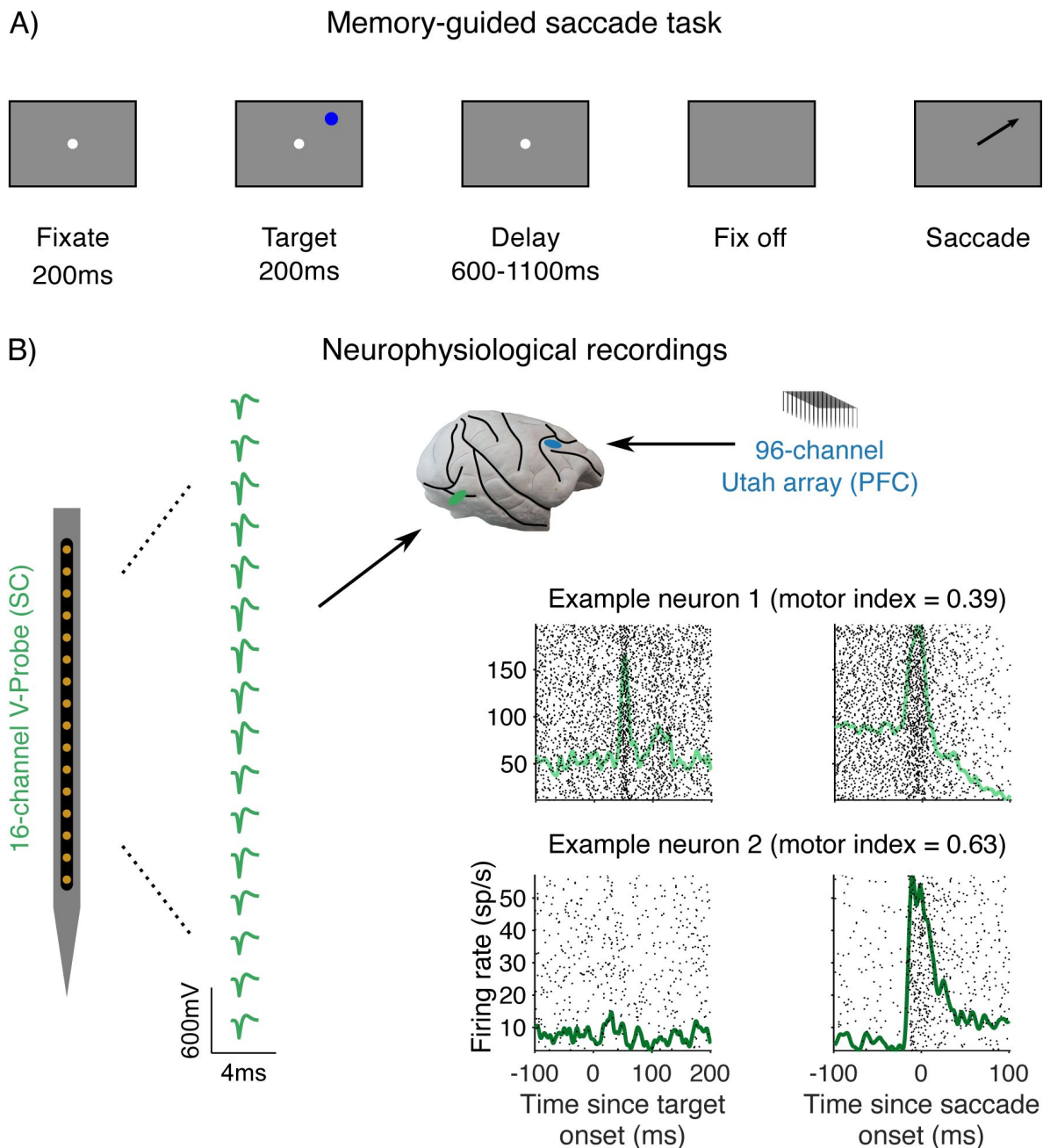


Figure 1.

Experimental methods.

(A) Memory-guided saccade task. After an initial fixation period, a target stimulus was presented at one of eight peripheral locations. This was followed by a delay period, the duration of which was chosen at random from a uniform distribution spanning 600-1100ms. The subjects' task was to remember the location of the target stimulus and make a saccade to it when the fixation point was extinguished. (B) Neurophysiological recordings. The spiking responses of neuronal populations in the SC were recorded for targets that were visually identified to be within the neuron's RF by applying low-amplitude microstimulation to contacts along the linear microelectrode array to evoke saccades (see Methods). The 16-channel linear microelectrode array was lowered into the brain through a recording chamber and the mean waveform for each channel during a single recording session is shown alongside a schematic of the linear array for one session. PSTHs aligned to target onset (left) and saccade onset (right) were computed to determine the response properties of individual SC neurons. Firing rates were computed in 1ms bins, averaged across trials, and smoothed using a Gaussian function for visualization purposes only ($\sigma = 5\text{ms}$).

Wurtz, 1983 [\(Figure 1A\)](#). For each neuron, peri-stimulus time histograms (PSTHs) were computed relative to: 1) the onset of the target stimulus; and 2) the onset of the saccade to the remembered target location ([Figure 1B](#)). We found that some of the neurons exhibited a visual response shortly after the presentation of the stimulus and a saccadic response when an eye movement was made to the remembered target location ([Figure 1B](#), Example neuron 1), traditionally termed visuomotor neurons, and were present mostly on the upper and middle recording contacts of our linear array. Others only exhibited a saccadic response ([Figure 1B](#), Example neuron 2), traditionally termed motor neurons, and were present mostly on the lower recording contacts of our linear array. Based on what is known about the response properties of neurons in the SC, these findings suggest that our linear array recordings targeted neurons residing at the mid to late stages of the visuomotor transformation i.e. the intermediate and deep layers of the SC (Basso and May, 2017 [\(Figure 1 - figure supplement 1\)](#)).

To quantify the observations we made about the strength of the saccadic response, we computed a motor index for each neuron (see Methods). SC neurons with a higher motor index (i.e. those with a stronger saccadic response relative to a baseline) were assumed to reside closer to the motor output stage of the visuomotor transformation than neurons with a lower motor index. Across sessions, we found that the mean motor index for all recorded SC neurons was 0.2761 (SD = 0.1924) ([Figure 1 - figure supplement 1](#)). Neurons recorded on deeper contacts tended to have a higher motor index in most sessions (see [Figure 3A](#) for an example), though our sampling was not equal across sessions and in some cases we found mostly visuomotor or mostly motor neurons across the responsive contacts on the linear array.

Arousal-related fluctuations are present in the SC and correlated with pupil size and fluctuations in PFC activity

We first asked if arousal-related fluctuations are present in the SC. As in previous studies that recorded from neurons in the cortex (Cowley et al., 2020 [\(Figure 2A\)](#)), we found that the mean spiking responses of individual SC neurons during the delay period (chosen at random on each trial from a uniform distribution spanning 600-1100ms, see Methods) fluctuated over the course of a session while the monkeys performed the MGS task ([Figure 2A](#), left). Note that this effect cannot be attributed to signals related to stimulus or saccade tuning as residual spike counts were computed by subtracting the mean response to each target location from trial-to-trial responses to each target location. These fluctuations are also not likely to reflect instabilities in our linear array recordings, as several steps were taken to remove neurons that exhibited changes in waveform shape over time or large sudden changes in activity level (see Methods).

To identify coordinated fluctuations in activity across the population of neurons, we applied PCA to residual spike counts of individual neurons during the delay period that had been binned using a 15-min sliding window stepped every 3 mins. This yielded a vector of loadings ([Figure 2A](#), middle) that we termed the “slow drift axis” in line with previous studies (Cowley et al., 2020 [\(Figure 2A\)](#)). The value of each loading corresponds to the correlation coefficient between the binned response of each SC neuron and the axis in population activity space that explained the most variance in the data. Thus, neurons whose residual activity fluctuated more over the course of a recording session had a greater portion of their variance explained by the slow drift axis (e.g. [Figure 2A](#), Channel 2). After we identified this axis from the binned data, we then projected the residual responses of each SC neuron on single trials, which resulted in a single dimension along which we examined fluctuations over time ([Figure 2A](#), bottom right). When the data were pooled across sessions, we found that the median proportion of variance explained by the slow drift axis was 0.88 ([Figure 2A](#), top right). By comparing the actual distribution of values to a null distribution that was generated by shuffling the trial-to-trial spike counts for each neuron prior to performing PCA, we were able to show that the variance explained by the slow drift axis was significantly greater than what would be expected by chance ($p = 0.0385$). Note also that the dynamics of the SC slow drift were highly similar when it was computed using spiking responses during the baseline, visual and

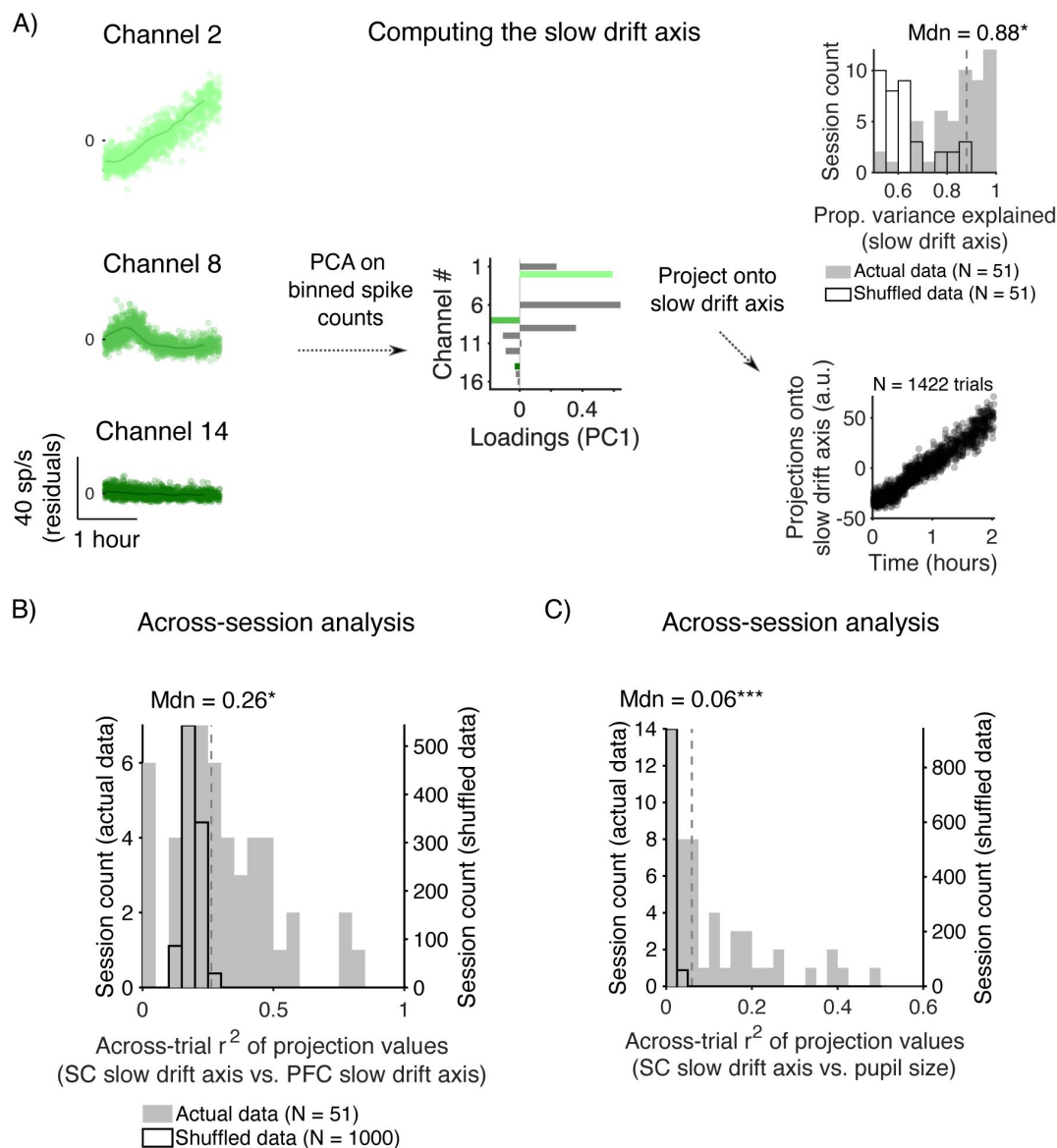


Figure 2.

Arousal-related fluctuations are present in the SC and correlated with pupil size and fluctuations in PFC activity.

(A) Computing the slow drift axis. Three example SC neurons from the same recording session (left). Each point represents the mean residual firing rate during the delay period. When PCA was applied to binned spike counts (solid line overlaid on each neuron's spike count residuals) it yielded a vector of loadings for the first principal component, which we will term the "slow drift axis" (middle). trial-to-trial residual SC responses during the delay period were then projected onto this vector (bottom right). The histogram in the top right shows the proportion of variance explained by the slow drift axis across sessions. The median value is indicated by the vertical dashed line overlaid on the histogram. A p-value was computed by comparing the actual distribution of values to a shuffled distribution (see Methods). (B) Histogram showing the proportion of variance explained between projections onto the SC slow drift axis and projections onto the PFC slow drift axis. Note that in this analysis, the slow drift axis was computed separately for each region using the method shown in (A). Trial-to-trial residual SC responses during the delay period and trial-to-trial residual PFC responses were then projected onto the SC slow drift axis and the PFC slow drift axis, respectively. The median value across sessions is denoted by the vertical dashed line overlaid on the histogram. A p-value was computed by comparing the actual distribution of values to a shuffled distribution (see Methods). (C) Same as (B) but in this case the histogram shows the proportion of variance explained between projections onto the slow drift axis and pupil size. $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

saccade epochs (**Figure 2 - figure supplement 1** [↗](#)). These findings suggest that arousal-related fluctuations are present throughout different task epochs and are less evident in deep-layer SC neurons independent of the time window in which spiking responses were computed.

To investigate if the SC slow drift axis was related to a global signal that might be related to shifts in arousal, we computed an identical axis using simultaneously recorded PFC data. If the SC slow drift axis represented an arousal axis that reflected coordinated activity fluctuations throughout the brain, one would expect it to be correlated with simultaneously recorded data in the PFC. For each session, we computed the correlation (Pearson's product-moment correlation coefficient) between projections onto the SC slow drift axis and projections onto the PFC slow drift axis. As shown in **Figure 2B** [↗](#), we found that the median variance explained across sessions, as measured using the coefficient of determination (r^2), was 0.2627. To determine whether or not this effect was statistically significant, we compared the actual distribution of r^2 values to a null distribution that was generated by shuffling across sessions and recomputing the correlation between projection values. We found that the proportion of variance explained was significantly greater than what would be expected if there was no correlation between projections onto the SC slow drift axis and the PFC slow drift axis ($p = 0.0190$). These findings suggest that slow fluctuations in responsivity are present in populations of SC neurons, and furthermore that they are coordinated across the brain. To further explore how they relate to changes in arousal or brain state, we next asked if the SC slow drift axis is associated with eye metrics that are known to be correlated with arousal levels.

Decades of research has shown that the size of the pupil can be used as a reliable marker of arousal (Mathôt, 2018 [↗](#); Joshi and Gold, 2020 [↗](#)). Heightened levels of arousal are accompanied by concomitant increases in LC activity and pupil size (Aston-Jones and Cohen, 2005 [↗](#); Murphy et al., 2014 [↗](#); Joshi et al., 2016 [↗](#); Reimer et al., 2016 [↗](#); de Gee et al., 2017 [↗](#); Breton-Provencher and Sur, 2019 [↗](#)). Thus, we next sought to explore if projections onto the SC slow drift axis were associated with pupil size. For each session, we computed the correlation (Pearson's product-moment correlation coefficient) between projections onto the SC slow drift axis and mean pupil size during the first 200ms of the delay period when a task-related pupil response could be observed. Across sessions, we found that the median variance explained between projections onto the SC slow drift axis and pupil size was 0.0597 (**Figure 2C** [↗](#)). When the actual distribution of values was compared to an across-session shuffle distribution, we found that the proportion of variance explained was significantly greater than what would be expected if there was no correlation between SC slow drift and pupil size ($p < 0.001$). Interestingly, the median variance explained between projections onto the SC slow drift axis and other eye metrics including microsaccade rate was much smaller ($r^2 = 0.0026$, $p < 0.001$) (**Figure 2 - figure supplement 2** [↗](#)). Although microsaccade rate (i.e. the number of small eye movements made during a period of steady fixation) has been linked to arousal (Siegenthaler et al., 2014; Johnston et al., 2022a [↗](#); Contadini-Wright et al., 2023 [↗](#)), microsaccade rates are task-dependent (Winterson and Collewyn, 1976 [↗](#); Ko et al., 2010 [↗](#); Bowers et al., 2021 [↗](#)) and may be driven by activity in cortical as opposed to subcortical regions (Hafed et al., 2021 [↗](#)). This might explain why no significant correlation was found between SC slow drift and microsaccade rate in the present study.

Taken together, these findings point to a link between the SC slow drift axis and brain-wide arousal signals. That simultaneously recorded PFC data and pupil size were correlated with projections onto the slow drift axis suggests that the signal we identified in the SC is highly similar to that which has been observed across several cortical areas (Cowley et al., 2020 [↗](#); Hennig et al., 2021 [↗](#)). The fact that signals related to arousal are present in the SC is not particularly surprising given that its functional role has evolved and is now thought to include higher order cognitive processes (Knudsen, 2011 [↗](#); Krauzlis et al., 2013 [↗](#); Clark et al., 2015 [↗](#); Basso et al., 2021 [↗](#)). However, it also poses a problem. Given the well-established role of deep-layers SC neurons in saccade generation (Sparks and Hartwich-Young, 1989 [↗](#); Gandhi and Katnani, 2011 [↗](#)), having signals related to cognition and arousal embedded in oculomotor circuits that send gaze

commands to downstream nuclei could lead to unwanted changes in eye position. One possibility is that arousal-related fluctuations are isolated from these deep-layer SC neurons that reside closer to the motor output. If this is the case, one would expect neurons with a stronger saccadic response to minimally fluctuate over time, and have a smaller portion of their variance explained by fluctuations along the slow drift axis.

Arousal-related fluctuations are isolated from SC neurons with a higher motor index

To investigate if arousal-related fluctuations are isolated from deep-layer SC neurons, we took advantage of the fact that our linear array recordings targeted neurons residing in earlier and later stages of the visuomotor transformation (**Figure 1B**). As described above, this was quantified by computing a motor index for each SC neuron with high values indicating that a cell elicited a strong saccadic response to the remembered target location and low values indicating that a cell elicited a weak saccadic response. Cells with a higher motor index tend to reside in the deep layers of the SC closer to the motor output (Basso and May, 2017). In several sessions, such as that shown in **Figure 3A**, we observed that the loadings for the slow drift axis (i.e. the correlation between the response of each neuron and the component that explained the most variance in the data) were weaker for neurons with a higher motor index. Of note in this particular session is that these neurons were recorded from channels at the bottom of the probe, which presumably targeted the deep layers of the SC.

To explore if a similar effect was found across sessions, we computed the squared component loading for all recorded SC neurons. This was done to control for the fact that the sign of the correlation between the response of each neuron and the slow drift axis is arbitrary and could be positive or negative (Jolliffe and Cadima, 2016). Next, we discretized the motor indices of all recorded SC neurons into four equally spaced bins. Permutation tests were then performed across the four bins to investigate if neurons with a stronger saccadic response explained less variance in the slow drift axis. After controlling for multiple comparisons (Bonferroni correction), we found that the mean squared component loading for neurons with the highest motor index ($M = 0.0105$, $SD = 0.0237$) (i.e. the bin centered on 0.75) was significantly weaker than the mean loading for neurons with lower motor indices (bin centered on 0.25: $M = 0.1539$, $SD = 0.2207$; bin centered on 0: $M = 0.1743$, $SD = 0.2429$) (**Figure 3B**). Note that similar results were obtained when motor index was entered as a continuous variable into a multiple regression analysis instead of being discretized into four equally spaced bins (**Figure 1 - figure supplement 1**). The proportion of variance explained by the model, which also included subject name as a categorical variable to control for intersubject differences, was 0.0350 ($F(2, 368) = 6.6748$, $p = 0.0014$). Consistent with our hypothesis, we found that motor index was significantly associated with squared component loadings ($t = -3.6154$, $p = 0.0003$) such that SC neurons with a higher motor index explained less variance in the slow drift axis. These results demonstrate that even though arousal-related fluctuations are present in the SC, they are isolated from deep-layer neurons that elicit a strong saccadic response and presumably reside closer to the motor output. As a further test of this hypothesis, we next sought to determine if these same cells explained less variance in projections onto the PFC slow drift axis.

SC neurons with a higher motor index are less correlated with the PFC slow drift axis

To probe the relationship between deep-layer SC neurons and the PFC slow drift axis, we computed the correlation (Pearson's product-moment correlation coefficient) between the trial-to-trial residual responses of each SC neuron during the delay period and projections onto the PFC slow drift axis. Consistent with our finding that SC slow drift was significantly correlated with PFC slow drift (**Figure 2B**), we observed that the trial-to-trial residual responses of some SC neurons were significantly correlated with projections onto the PFC slow drift axis. For example, the

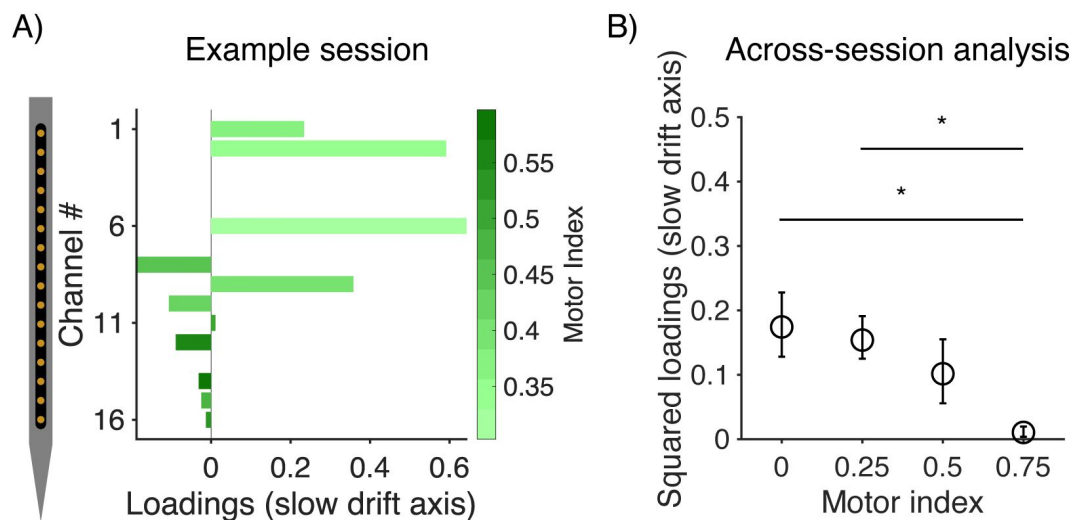


Figure 3.

Arousal-related fluctuations are isolated from SC neurons with a higher motor index.

(A) Loadings for the slow drift axis for an example session. Motor index is represented by the colorbar. Note that loadings were weaker for neurons recorded from channels at the bottom of the probe, which targeted neurons in the deep layers of the SC. These neurons also had higher motor indices, consistent with their role in eye movement generation. (B) Scatter plot for the across session analysis showing squared loadings for the slow drift axis pooled across all recorded SC neurons. The neurons were discretized into four equally spaced bins based on motor index, and permutation tests performed to investigate if neurons with a higher motor index explained less variance in the slow drift axis. The resultant p-values were then corrected for multiple comparisons using Bonferroni adjustment. $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

residual responses of the neuron shown in **Figure 4A**, which has a relatively low motor index and is therefore unlikely to reside close to the motor output of the SC, were strongly correlated with projections onto the PFC slow drift axis ($r^2 = 0.5346$, $p < 0.001$). However, other neurons showed no correlation such as that in **Figure 4B**. The residual responses of this neuron, which has a relatively high motor index and presumably resides closer to the motor output stage, were not significantly correlated with projections onto the PFC slow drift axis ($r^2 = 0.0010$, $p = 0.2650$).

To explore if a similar effect was found across sessions, we again pooled the motor indices of all recorded SC neurons and discretized them into four equally spaced bins. We performed permutation tests across the bins to investigate if the mean variance explained between residual SC neuron responses and projections onto the PFC slow drift axis was lower for neurons with a higher motor index. After controlling for multiple comparisons, we found that the mean variance explained was significantly lower for neurons with the highest motor index SC ($M = 0.0209$, $SD = 0.0236$) (i.e. the bin centered on 0.75) than it was for neurons with lower motor indices (bin centered on 0.25: $M = 0.1240$, $SD = 0.1468$; bin centered on 0: $M = 0.1557$, $SD = 0.1591$) (**Figure 4C**). Furthermore, the mean variance explained for neurons with an intermediate motor index (i.e. the bin centered on 0.5) ($M = 0.0703$, $SD = 0.1095$) was significantly lower than it was for neurons with the lowest motor indices (i.e. the bins centered on 0.25 and 0). Once again, similar results were obtained when the motor index was entered as a continuous variable into a multiple regression model instead of being discretized into bins (**Figure 1 - figure supplement 1**). The proportion of variance explained by the model, after controlling intersubject differences, was 0.1777 ($F(2, 368) = 39.7739$; $p < 0.001$). Motor index was significantly associated with the strength of the relationship between single SC neurons responses and projections onto the PFC slow drift axis ($t = -2.7902$; $p = 0.0055$) such that it was weaker for neurons with a higher motor index. These findings provide additional support for the hypothesis that arousal-related fluctuations are isolated from neurons in the deep layers of the SC. They are more characteristic of cells residing at the early to mid stages of the visuomotor transformation, which have been implicated in autonomic processes such as controlling the size of the pupil (Wang et al., 2012; Wang and Munoz, 2018). This prompted us to ask if a relationship exists between SC neurons responses and pupil size, and whether this relationship is mediated by motor index.

The relationship between SC neuron responses and pupil size is modulated by motor index

Previous work has shown that intermediate-layer SC neurons control the size of the pupil via: 1) direct projections to the Edinger-Westphal nucleus; and 2) indirect projections to the mesencephalic cuneiform nucleus (Joshi and Gold, 2020). Thus, we hypothesized that neurons with a lower motor index would explain more variance in trial-to-trial pupil size than neurons in deep layers. To test this, we computed the correlation (Pearson's product-moment correlation coefficient) between the spiking responses of each SC neuron and pupil size, both of which were computed during the first 200ms of the delay period. In several sessions, we found that the strength of this relationship was inversely related to motor index. For example, the spiking response of the neuron shown in **Figure 5A**, which has a relatively low motor index and presumably resides at the mid stage of the visuomotor transformation, was strongly correlated with pupil size ($r^2 = 0.2121$, $p < 0.001$). In contrast, the spiking responses of the neuron shown in **Figure 5B**, which has a relatively high motor index and presumably resides closer to the motor output stage, was not significantly correlated with changes in pupil size over the course of a recording session ($r^2 = 0.001$, $p = 0.7534$).

To determine if a similar effect was found across sessions, we performed a multiple regression analysis. In this analysis, we sought to determine if motor index predicted the strength of the relationship between SC neuron responses during the delay period and pupil size, as measured using the coefficient of determination (r^2). After controlling for intersubject differences, results showed that the proportion of variance explained by the model was 0.0309 ($F(2, 368) = 5.8717$; $p =$

Figure 4.

SC neurons with a higher motor index are less correlated with the PFC slow drift axis.

(A) Example SC neuron with a relatively low motor index that exhibited a significant correlation between its residual spiking response and projections onto the PFC slow drift axis ($r^2 = 0.5346$, $p < 0.001$). (B) Example SC neuron with a relatively high motor index that did not exhibit a correlation between its residual spiking response and projections onto the PFC slow drift axis ($r^2 = 0.001$, $p = 0.2650$). (C) Scatter plot for the across session analysis showing the mean variance explained between residual spiking responses and projections onto the PFC slow drift axis for all recorded SC neurons. The neurons were discretized into four equally spaced bins based on motor index, and permutation tests performed to investigate if the variance explained between residual SC spiking responses and projections onto the PFC slow drift was lower for neurons with a higher motor index. The resultant p-values were then corrected for multiple comparisons using Bonferroni adjustment. $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

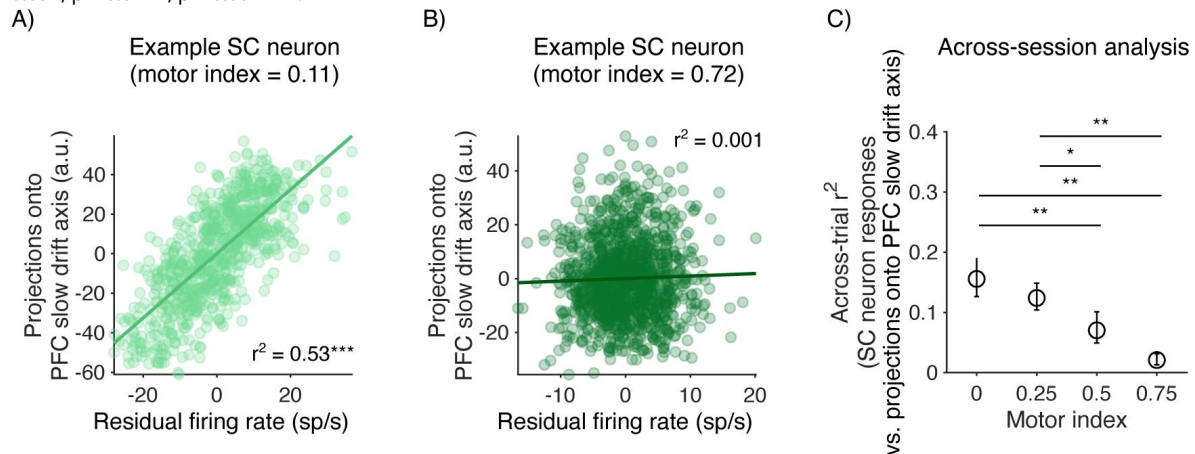
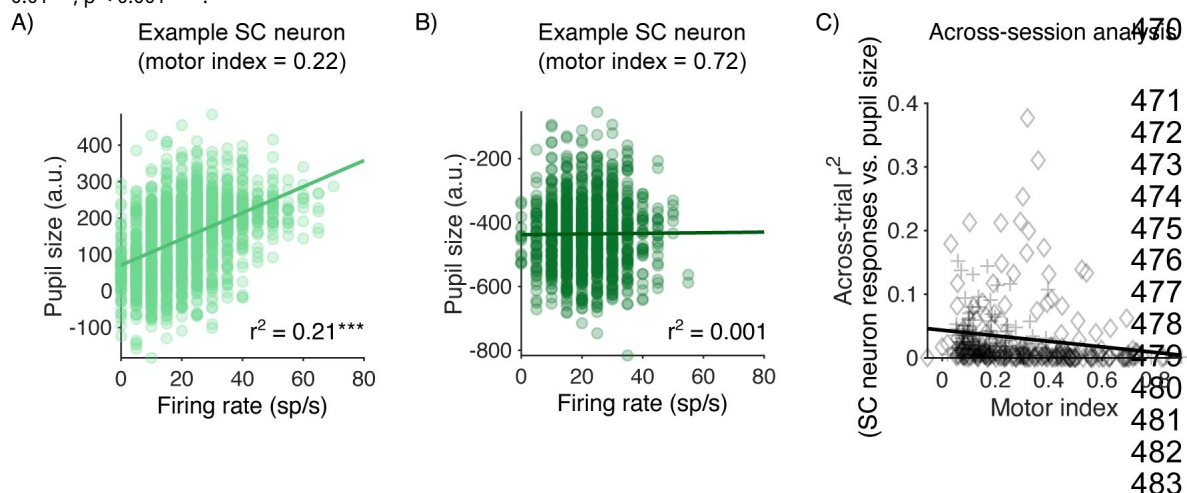


Figure 5.

The relationship between SC neuron responses and pupil size is modulated by motor index.

(A) Example SC neuron with a relatively low motor index that exhibited a significant correlation between its spiking response and pupil size. (B) Example SC neuron with a relatively high motor index that did not exhibit a correlation between its spiking response and pupil size ($r^2 = 0.1601$, $p < 0.001$). (C) Scatter plot showing that the strength of the relationship between individual SC neuron responses and pupil size is inversely related to motor index ($r^2 = 0.0309$, $p = 0.0031$). $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.



0.0031). Motor index was significantly associated with the strength of the relationship between SC neuron responses and pupil size ($t = -3.2593$; $p = 0.0012$) (**Figure 5C** [↗](#)). More specifically, we found a stronger relationship between neuronal response and pupil size for neurons with a lower motor index, consistent with the proposed role of intermediate-layer SC neurons in controlling the size of the pupil (Joshi and Gold, 2020 [↗](#)).

We also considered whether shifts in SC neural activity could be driven directly by changes in luminance induced by changes in the size of the pupil. If such luminance sensitivity existed in SC among the visually selective neurons, it could explain the slow fluctuations in their activity correlating with slow fluctuations in the size of the pupil. To ask whether this might be the case, we performed two analyses. First, we analyzed the pupil size in a different period of time, the baseline period before target presentation. Although there was only a fixation point on the screen at that time and no target to induce a transient visual response to luminance, we found similar results relating SC spiking responses to pupil size (**Figure 5 - figure supplement 1** [↗](#)). Second, we compared slow drift measures computed on spiking responses during different epochs, and found they were highly correlated (**FIGURE 2 - figure supplement 1** [↗](#)). Because slow drift is computed on an average spiking across many trials, this makes it unlikely that transient luminance-induced responses could lead to changes in these fluctuations. In our previous work, we also found that slow drift occurred on a timescale of minutes in conjunction with changes in pupil size (Cowley et al., 2020 [↗](#)) and were correlated with eye measures such as reaction time, saccadic velocity and microsaccade rate (Johnston et al., 2022a [↗](#)). Together, these analyses and previous results make luminance changes induced by shifts in pupil size an unlikely explanation for slow fluctuations in the firing of visually-responsive neurons in SC.

Taken together, the findings outlined above suggest that arousal-related signals are kept separate from neurons in the deep layers of the SC. In addition to being more weakly correlated with pupil size, the spiking responses of these neurons did not exhibit large fluctuations over time (**Figure 2** [↗](#)), and when considering the neuronal population as a whole, explained less variance in the slow drift axis when it was computed using population activity in the SC (**Figure 3** [↗](#)) and PFC (**Figure 4** [↗](#)). Although these results point to a segregation of arousal- and eye movement-related signals in the SC, research has shown that all three layers receive input from the LC: a brain region that has long been associated with controlling global arousal levels (Benavidez et al., 2021 [↗](#)). Thus, an additional mechanism may have evolved to ensure that arousal-related signals are kept separate from those used in the encoding of eye position. In primary motor cortex, signals related to movement preparation reside in an orthogonal axis in population activity space to those that are used for movement initiation so that they do not lead to premature muscle contractions (Kaufman et al., 2014 [↗](#); Elsayed et al., 2016 [↗](#)). Next, we explored if a similar mechanism exists in the SC to prevent arousal-related signals interfering with the motor output.

Arousal-related signals in the SC reside in an orthogonal subspace to saccade-related signals

To investigate if arousal-related signals in the SC reside in an orthogonal subspace to those that are used to encode eye position, we computed a “pupil size axis” for each session. We discretized trial-to-trial pupil size into eight equally spaced bins and, for each neuron, computed the mean spiking response per bin (**Figure 6A** [↗](#), left). We then used PCA to find the axis in population activity space that explained the most variance in the data attributable to the pupil. The loadings (**Figure 6A** [↗](#), right) represent the correlation between the response of each neuron across bins and the first principal component meaning that neurons exhibiting a stronger relationship between firing rate and pupil size (e.g. **Figure 6A** [↗](#), Channel 5) explained more variance in the pupil size axis. Our goal was to compare projections onto the pupil size axis to projections onto a “saccade tuning axis”. To compute this axis, tuning curves were generated for each SC neuron by taking the mean spiking response during the saccade epoch for each target angle (**Figure 6B** [↗](#), right). PCA was then applied to the tuning curves, which yielded a vector of loadings for the first principal

component (**Figure 6B**, left). In this case, neurons that exhibited greater selectivity for a particular target angle explained more variance in the saccade tuning axis (e.g. **Figure 6B**, Channel 11).

We first investigated if the pupil size axis and the saccade tuning axis reside in orthogonal subspaces. To ensure that fair comparisons could be made, the same data (i.e. trial-to-trial residual responses during the delay period) were projected onto each of the axes. We then computed a correlation (Pearson's product-moment correlation coefficient) between the projection values for each session, and used the coefficient of determination (r^2) to assess the variance explained. If arousal-related signals in the SC reside in an orthogonal subspace to those that are used for saccade generation, one would expect projections onto the pupil size axis to be weakly correlated with projections onto the saccade tuning axis. This was found to be the case in several sessions such as that shown in **Figure 6C** ($r^2 = 0.0307$, $p < 0.001$). Furthermore, when the data were pooled across sessions, we found that the mean variance explained between the projection values was 0.5142 (**Figure 6D**). By comparing the actual distribution of r^2 values to a null distribution that was generated by correlating projections onto the pupil size axis with projections onto an axis that was randomly oriented in population activity space, we were able to show that the mean variance explained was not significantly greater than what would be expected for data projected onto a random axis ($p = 0.5192$).

Finally, as an additional test of our hypothesis, we investigated if the pupil size axis was more aligned to the slow drift axis than the saccade tuning axis. To do this, we projected trial-to-trial residual responses during the delay period onto the pupil size axis and the saccade tuning axis. For each session, we then correlated (Pearson's product-moment correlation coefficient) these projection values with projections onto the slow drift axis. Across sessions, we found that the mean variance explained between projections onto the pupil size axis and the slow drift axis ($M = 0.7780$, $SD = 0.3025$) was significantly greater than the mean variance explained between projections onto the saccade tuning axis ($M = 0.5319$, $SD = 0.3407$) ($t = -4.8578$, $p < 0.001$).

Discussion

The main aim of this study was to identify how global fluctuations in arousal affect subcortical areas like the SC, and how such fluctuations interact with the motor functions of that area. In addition to representing a highly conserved brain structure in vertebrates, evidence suggests that the SC also has a number of divergent properties that have been linked to higher-order cognitive processes such as spatial attention (Knudsen, 2011; Krauzlis et al., 2013; Clark et al., 2015; Basso et al., 2021) and perceptual decision-making (Ratcliff et al., 2003; Horwitz et al., 2004; Crapse et al., 2018; Jun et al., 2021). In a similar way to how cortical motor areas separate out preparatory signals (Kaufman et al., 2014; Elsayed et al., 2016) and signals from the opposing arm (Ames and Churchland, 2019) when performing a reach movement, oculomotor structures such as the SC must be able to disentangle signals related to cognition and brain state from those used for eye movement encoding.

Consistent with what has been observed in the cortex (Cowley et al., 2020), we found that the activity of some SC neurons fluctuated over time, and by reducing the dimensionality of the data we were able to isolate a neuronal signature of arousal that was correlated with pupil size and simultaneously recorded PFC data. In the context of the SC, this motivated us to ask how the precise coordination of saccades is maintained despite the presence of ongoing fluctuations in brain state. Another internal cognitive state that has been studied in the context of the visual and motor role of the SC is spatial attention. Research using single-unit recordings in monkeys (Ignashchenkova et al., 2004) and neuroimaging in humans (Katyal et al., 2010) has shown that the effects of spatial attention on SC activity are layer-specific and more characteristic of neurons in the superficial and intermediate layers. At a population level, attention has been

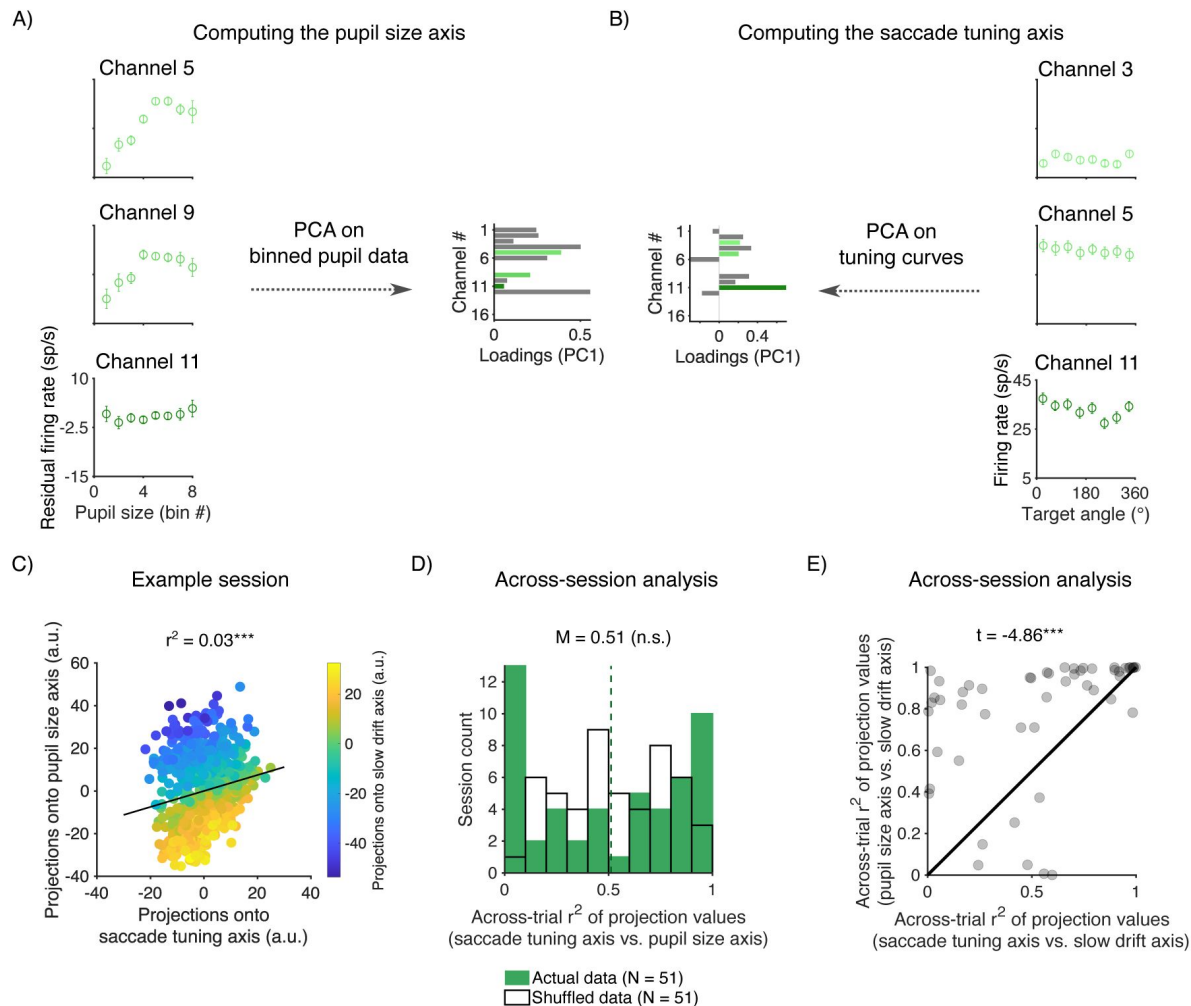


Figure 6.

Arousal-related signals in the SC reside in an orthogonal subspace to saccade-related signals.

(A) Computing the pupil size axis for an example session. Three example SC neurons from the same recording session (left). Each data point represents the mean spiking response after trial-to-trial pupil size had been discretized into eight equally spaced bins. Note that pupil size increases with bin number such that bin 1 contains trials in which the pupil was most constricted, whereas bin 8 contains trials in which the pupil was most dilated. When PCA was applied to the data it yielded a vector of loadings for the first principal component (right). (B) Computing the saccade tuning axis for the same session shown in (A). Three example SC neurons (right). Each point represents the mean spiking response during the saccade epoch (25ms before to 25ms after the onset of the saccade) for each target angle. When PCA was applied to the data it yielded a vector of loadings for the first principal component (left). (C) A scatter plot showing the relationship between projections onto the pupil size axis and the saccade tuning axis for the same example session shown in (A) and (B). The color scale represents projections onto the slow drift axis. Note that the same data (i.e. residual SC responses during the delay period) were projected onto the three axes. (D) Histogram showing the proportion of variance explained between projections onto the pupil size axis and projections onto the saccade tuning axis. A p-value was computed by comparing the actual distribution of values to a null distribution (see Methods). (E) Scatter plot comparing the variance explained between projections onto the pupil size axis and the slow drift axis, and projections onto the saccade tuning axis and the slow drift axis. A p-value was computed using a paired-sample t-test. $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

shown to increase the efficacy of communication from MT to SC in a similar manner for visual and motor SC populations (Srinath et al., 2021 [DOI](#)) and reshape the visual information provided by MT to SC (Ruff and Cohen, 2019 [DOI](#)). Overall, both spatial attention and global arousal share the property of having an influence on the eye movement system while at the same time needing to avoid directly producing unwanted eye movements. Here, we propose that arousal-related signals in the SC are segregated from the motor planning function of SC to prevent unwanted changes in eye position. To test this hypothesis, we utilized linear array recordings to simultaneously monitor the activity of SC neurons with distinct neurophysiological response properties (Massot et al., 2019 [DOI](#); Ayar et al., 2023 [DOI](#)). Across sessions, we found that neurons exhibiting a stronger saccadic response, which largely reside in the deep layers of the SC closer to the motor output (Basso and May, 2017 [DOI](#)), were less connected to SC slow drift measured in the population response. Furthermore, by correlating the response of these cells with a slow drift axis that was generated using simultaneously recorded PFC data, we were able to verify the global nature of this signal. These findings are consistent with neuroimaging studies showing that changes in brain state lead to coordinated fluctuations of activity across the brain (Liu et al., 2018 [DOI](#); Turchi et al., 2018 [DOI](#); Setzer et al., 2022 [DOI](#)), and research that recorded simultaneously from neurons in cortical areas and subcortical areas such as the LC (Joshi and Gold, 2022 [DOI](#); Collins et al., 2023 [DOI](#)). Furthermore, our results suggest that arousal-related fluctuations are more characteristic of neurons residing in the intermediate layers of the SC. In support of this, several studies have pointed to a link between the activity of these cells and non-invasive markers of arousal such as pupil size (Wang et al., 2012 [DOI](#), 2014 [DOI](#); Joshi et al., 2016 [DOI](#); Wang and Munoz, 2021 [DOI](#)).

To further study the role of intermediate-layer SC neurons in controlling the size of the pupil, we investigated if the strength of the relationship between trial-to-trial changes in pupil size and SC spiking responses was mediated by motor index. Consistent with the proposed link between intermediate-layer neurons and pupil size, we found that cells exhibiting a weaker saccadic response explained more variance in pupil size than neurons with a stronger saccadic response. These results provide further support for the notion that arousal-related signals are kept separate from signals used to encode eye position in the SC. This provides a potential mechanism through which signals related to cognition and brain state can exist in oculomotor structures without affecting high precision fixation and eye movements.

In addition to being kept separate at the level of different layers in the SC, signals related to arousal and eye position encoding may also occupy different dimensions in population activity space. The term “subspace” is often used when referring to the fact that the activity of simultaneously recorded neurons is correlated (Cohen and Kohn, 2011 [DOI](#)) and occupies fewer dimensions than the number of neurons in the population (Ebitz and Hayden, 2021 [DOI](#); Rust and Cohen, 2022 [DOI](#)). The concept of a subspace has been used to identify how populations of neurons segregate different types of activity at the population level, rather than simply in different neurons. Non-overlapping subspaces can be used to encode different types of sensory information (Mante et al., 2013 [DOI](#); Aoi et al., 2020 [DOI](#); Ebitz et al., 2020 [DOI](#)), facilitate communication between distinct neural circuits (Semedo et al., 2019 [DOI](#), 2022 [DOI](#)), and isolate preparatory signals from movement commands (Churchland and Shenoy, 2024 [DOI](#)). For example, when recording from neurons in the motor cortex while monkeys performed a delayed-reach task, (Kaufman et al., 2014 [DOI](#)) found that preparatory signals reside in an orthogonal subspace to those used to encode reach direction. In the SC, (Ayar et al., 2023 [DOI](#)) found that signals related to sensation and action occupy separate subspaces and depend upon the cognitive demands of the task, consistent with previous research showing that the temporal structure of the SC population response differs during visual and motor epochs (Jagadisan and Gandhi, 2022 [DOI](#)). To explore if a similar mechanism exists in the SC, we projected trial-to-trial responses onto a pupil size axis and a saccade encoding axis, and found that the variance explained was not significantly greater than what would be expected for data projected onto a random axis. Furthermore, projections onto the pupil size axis were more aligned to projections onto the slow drift axis than projections onto the saccade encoding axis. Taken together, these findings suggest that arousal-related signals reside in

an orthogonal subspace to those used to encode eye movement plans. In a similar manner to how preparatory signals in the motor cortex do not lead to premature muscle contractions (Kaufman et al., 2014; Elsayed et al., 2016), this might explain why arousal-related signals in the SC do not lead to unwanted changes in eye position.

In summary, by simultaneously recording from populations of neurons in the SC and the PFC, we were able to observe a fluctuation of activity that was present in both areas and correlated with non-invasive markers of arousal such as pupil size. The presence of arousal-related signals in the SC is consistent with its role in cognitive processes such as spatial attention and decision-making. However, it also raises the question of how motor areas can disentangle these higher-order signals from those used for movement encoding. In addition to being kept separate at the level of different layers in the SC, our results demonstrate that signals related to arousal and eye movement planning occupy distinct dimensions in population activity space. Further research is needed to determine whether this represents a similar mechanism to that which is used in the motor cortex to separate preparatory signals from movement commands (Churchland and Shenoy, 2024). If so, it could provide an explanation for how cognitive signals can be intermixed and impact activity in populations of SC neurons without affecting precision in fixation and saccadic eye movements.

Methods

Subjects

Two adult rhesus macaque monkeys (*Macaca mulatta*) were used in this study (Monkey Do = 28 sessions, Monkey Wa = 34 sessions). Surgical procedures to chronically implant a titanium head post (to immobilize the subjects' heads during experiments) and microelectrode arrays were conducted in aseptic conditions under isoflurane anesthesia. Recording chambers situated along the midline at an angle of 30° from vertical and aimed at a location 8mm above interaural zero were also implanted for SC recordings. Opiate analgesics were used to minimize pain and discomfort during the perioperative period. Experimental procedures were approved by the Institutional Animal Care and Use Committee of Carnegie Mellon University and were performed in accordance with the United States National Research Council's Guide for the Care and Use of Laboratory Animals.

Superior colliculus (SC) recordings

The spiking responses of neuronal populations in the SC were recorded using a 16-channel linear microelectrode array (V-Probe, Plexon, Dallas, TX). The total length of the linear array was 115mm. The contacts were spaced 150µm apart and spanned a distance of 2.25mm. Linear arrays were lowered into the SC through a guide tube using a custom-designed mechanical microdrive (Laboratory for Sensorimotor Research, Bethesda, MD), which was inserted through a plastic grid with 1mm spacing.

Prefrontal cortex (PFC) recordings

The spiking responses of neuronal populations in the PFC (Monkey Do = left hemisphere, Monkey Wa = right hemisphere) were recorded using 96-channel "Utah" arrays (Blackrock Microsystems). The arrays comprised a 10×10 grid of silicon microelectrodes (1mm in length) that were spaced 400µm apart, implanted on the pre-arcuate gyrus (8Ar).

Waveform classification

Signals from the linear probes (SC) and Utah arrays (PFC) were amplified and band-pass filtered (0.3–7500Hz) by a Grapevine system (Ripple). Waveform segments crossing a threshold (set as a multiple of the root mean square noise on each channel) were digitized (30KHz) and stored for

offline processing, which involved waveforms being automatically sorted as spikes or noise by a neural network classifier developed in our laboratory (code available at <https://github.com/smithlabvision/spikesort/nasnet>). Although this method cannot distinguish between multiple neurons recorded from the same channel, it provides fast and accurate classifications that are highly similar to those of a human spike-sorter (Issar et al., 2020). All waveforms recorded from a single channel that remained after the neural network classifier had been applied were considered to be from a single unit, which we refer to as a “neuron”.

Exclusion criteria

Across both monkeys, we recorded from 992 sites in SC and 5952 sites in PFC. This included all channels from the 16-channel linear microelectrode array and all channels from the 96-channel Utah array across all sessions (SC: Monkey Do = 448, Monkey Wa = 544; PFC: Monkey Do = 2688, Monkey Wa = 3264). We applied a rigorous set of exclusion criteria designed to eliminate neurons with noisy or unstable waveforms, those with low firing rates, and neurons that did not have an evoked response during either the visual or motor epoch of the MGS.

To identify candidate neurons that were mostly classified as containing noise waveforms by the neural network classifier, and therefore only comprised a small number of waveforms, we removed neurons that had a mean firing rate less than 2sp/s across the entire session (Smith and Sommer, 2013). This was calculated by taking the total number of spikes in the recording session from the beginning (before the task was started) to the end (after the last trial of the day), and dividing by the total time (including periods of task performance and periods outside of the trial). In addition to removing neurons that had a low mean firing rate across an entire session, we also removed SC neurons that did not exhibit elevated firing rates during the visual and motor epochs of the MGS tasks i.e. following the presentation of the target stimulus and during the onset of a saccade. This was quantified by computing a visual and motor index for each SC neuron (see below). Neurons were removed if the value of both of these indices was less than 0.05, indicating a lack of response during both epochs. To rule out the possibility that slow drift arose due to recording instabilities (e.g. the distance between the neuron and the microelectrode(s) changing over time), we only included neurons with stable waveform shapes throughout a session. Neurons were removed if they had a percent waveform variance greater than 10% (Cowley et al., 2020). We also removed neurons that exhibited abrupt changes in activity during a recording session. This was quantified by dividing each session into 20 non-overlapping time bins and computing the mean spike count per bin (Cowley et al., 2020). Neurons were removed if the change in activity between any two consecutive time bins was greater than 25% of the maximum activity of that neuron. Adopting these exclusion criteria ensured that slow drift did not arise due to recording instabilities. We also note that if drift were to occur due to recording instabilities, it would not be expected to affect neurons from the same penetration differentially based on their motor index (e.g., Figure 3A).

In order to obtain a precise estimate of the slow drift axis, and after controlling for low firing rate and recording instabilities, we removed sessions that contained less than two SC or PFC neurons (Monkey Do = 3 sessions, Monkey Wa = 5 sessions). Furthermore, we removed sessions if they had a duration lasting less than 45 minutes (Monkey Do = 0 sessions, Monkey Wa = 3 sessions). Adopting these additional criteria ensured that there were enough variables and observations to perform PCA and accurately compute the slow drift axis. Using all of this criteria, the total number of sessions included in the analysis for Monkey Do and Monkey Wa was 25 sessions (SC = 190 neurons, PFC = 795) and 26 sessions (SC = 181, PFC = 2110), respectively.

Visual stimuli

Visual stimuli were generated using a combination of custom software written in MATLAB (The MathWorks) and Psychophysics Toolbox extensions (Brainard, 1997; Pelli, 1997; Kleiner et al., 2007). They were displayed on a CRT monitor (resolution = 1024 X 768 pixels; refresh rate =

100Hz), which was viewed at a distance of 57cm and gamma-corrected to linearize the relationship between input voltage and output luminance using a photometer and look-up-tables.

Memory-guided saccade task

Subjects fixated a central point (diameter = 0.18°) on the monitor to initiate a trial (**Figure 1A** [↗](#)). After fixating within a circular window (diameter: Monkey Do = 2.38°, Monkey Wa = 1.10°) for 200ms, a target stimulus (diameter = 0.55°) was presented for 200ms at one of eight angles separated by 45°. The eccentricity of the target stimulus was determined on a session-by-session basis by applying low-amplitude microstimulation to contacts along the linear microelectrode array to evoke saccadic eye movements. Target eccentricity was then set such that it matched the endpoints of evoked saccades. After the target stimulus had been presented, subjects were required to maintain fixation for a delay period. The duration of the delay period chosen at random on each trial from a uniform distribution spanning 600-1100ms. If steady fixation was maintained throughout the delay period, the central point was extinguished prompting the subjects to make a saccade to the remembered target location. The subjects had 500ms to initiate a saccade. To receive a liquid reward, the subjects' gaze had to be maintained within a circular window centered on the remembered target location (diameter: Monkey Do = 2.93°, Monkey Wa = 2.20° for the first 12 sessions and 3.01° for the remaining 22 sessions) for 100ms. In a subset of sessions, the target was briefly reilluminated, after the fixation point was extinguished and the saccade had been initiated, to aid in saccade completion.

Eye tracking

Eye position and pupil diameter were recorded monocularly at a rate of 1000Hz using an infrared eye tracker (EyeLink 1000, SR Research).

Microsaccade detection

Microsaccades were detected during the delay period and were defined as eye movements that exceeded a velocity threshold of 6 times the standard deviation of the median velocity for at least 6ms (Engbert and Kliegl, 2003 [↗](#)) (**Figure 2 - figure supplement 2** [↗](#)). To prevent overshoots (i.e. fast, oppositely directed gaze shifts that occur shortly after a microsaccade) being treated as separate eye movements, we merged microsaccades that occurred within 50ms of each other (McCamy et al., 2015 [↗](#)). Finally, we assessed the validity of our microsaccade detection method by computing the correlation (Pearson's product-moment correlation coefficient) between the amplitude and peak velocity of detected microsaccades for each session. The mean correlation between these two variables across sessions was 0.7601 (SD = 0.0678) indicating that our detection algorithm was robust, as microsaccades fell on the main sequence (Zuber et al., 1965 [↗](#)).

Characterizing the response properties of SC neurons

To characterize the neurophysiological response properties of individual SC neurons, we first visualized SC responses by computing PSTHs aligned to the onset of the target stimulus and the onset of the saccade to the remembered target location, respectively (**Figure 1B** [↗](#)). Firing rates were computed in 1ms bins, averaged across trials and smoothed using a Gaussian function ($\sigma = 5\text{ms}$).

To further quantify the strength of the saccadic response to the remembered target location, a motor index was computed for each neuron as follows:

$$\text{Motor index} = \frac{M - B}{M + B}$$

Where M represents the mean response during a peri-saccade epoch from 25ms before to 25ms after the onset of the saccade, and B represents the mean response during a baseline period defined as 100ms before the onset of the target stimulus. Higher motor index values are indicative

of a stronger saccadic response relative to baseline, whereas lower values represent a weaker saccadic response relative to baseline. Note that we also computed a visual index for each neuron, which was only used for the purpose of excluding neurons that did not exhibit an evoked response during the MGS task (see above). The visual index was computed in the following manner:

$$\text{Visual index} = \frac{V - B}{V + B}$$

Where V represents the mean response during a peri-stimulus epoch spanning 25ms to 125ms after the onset of the target stimulus, and B represents the mean response during a baseline period defined as 100ms before the onset of the target stimulus.

Generating shuffled distributions

Variance explained by SC slow drift axis across sessions

To determine if the variance explained by the SC slow drift axis was significantly greater than chance across sessions, the actual distribution of proportion variance explained values was compared to a shuffled distribution (**Figure 2A** [↗](#), top right). To generate this distribution, PCA was performed on binned spike counts that had first been shuffled across trials (1000 iterations). We then computed the mean variance explained by the first principal component across iterations, which yielded a single shuffled value for each session. Finally, a p-value was computed by calculating the proportion of values in the shuffled distribution that were greater than or equal to the actual variance explained across sessions.

Relationship between SC slow drift axis and the PFC slow drift

To assess the significance of the relationship between projections onto the SC slow drift axis and the PFC slow drift axis (**Figure 2B** [↗](#)), we randomly shuffled the two variables across sessions and re-computed the correlations (1000 iterations). Differences in session length were controlled for by truncating longer sessions such that they comprised the same number of data points as the shorter session. A p-value was then computed by calculating the proportion of values in the shuffled distribution that were greater than or equal to the actual variance explained across sessions. Note that an identical method was used to determine if projections onto the SC slow drift axis explained a significant amount of the variance in pupil size (**Figure 2C** [↗](#)) and microsaccade rate (**Figure 2 - figure supplement 2** [↗](#)).

Orthogonality of the pupil size axis and the saccade tuning axis

To determine if the pupil size axis and the saccade tuning axis reside in orthogonal subspaces, we computed the correlation between projections onto each axis and then compared the actual distribution of r^2 values across sessions to a shuffled distribution (**Figure 6D** [↗](#)). To generate this distribution, projections onto the pupil size axis were correlated with projections onto an axis that was randomly orientated in population activity space (1000 iterations). The latter was generated in a similar manner to the saccade tuning axis, the only difference being that tuning curves were computed for target angles that had been shuffled across trials. We then computed the mean variance explained across iterations, which yielded a single shuffled value for each session. Finally, a p-value was computed by calculating the proportion of values in the shuffled distribution that were greater than or equal to the actual variance explained across sessions.

Supplementary information

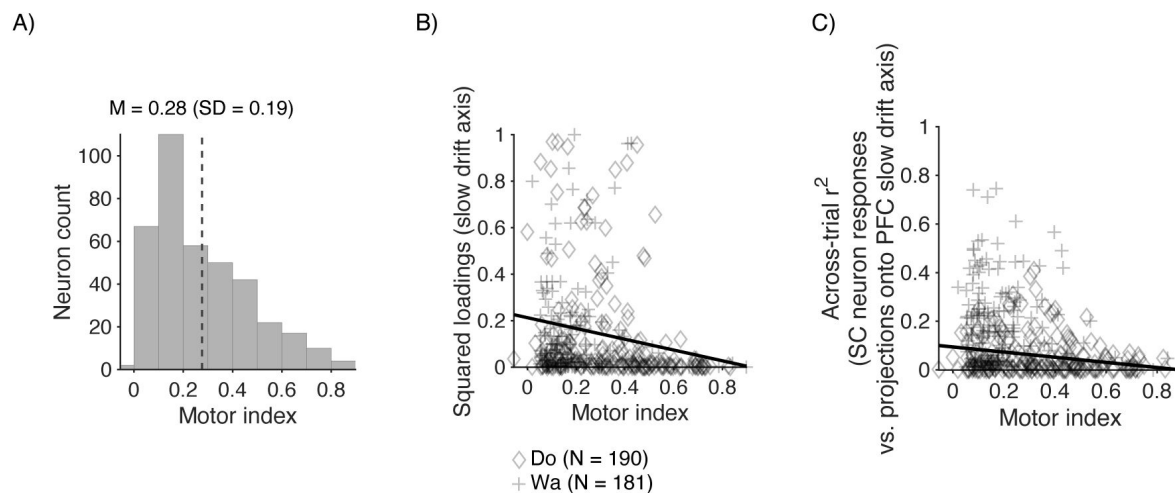


Figure 1 - figure supplement 1.

(A) Histogram showing the distribution of motor index values computed across sessions for all recorded SC neurons. The mean value across sessions is denoted by the dashed line. (B) Scatter plot showing the relationship between motor index and squared component loadings for the SC slow drift axis. The line of best fit was computed using multiple regression. Motor index was entered as a continuous predictor variable along with subject name to control for differences between subjects. As described in the Results section, the model explained a significant amount of the variance in the data. A negative correlation was also found between motor index and squared loadings. (C) Scatter plot showing the relationship between motor index and the strength of the relationship between single SC neurons responses and projections onto the PFC slow drift axis, as measured by computing r^2 . As in (B), the line of best fit was computed using multiple regression. Results showed that the model explained a significant amount of the variance in the data. Furthermore, a negative correlation was found between motor index and the r^2 between single neuron SC responses and projections onto the PFC slow drift axis.

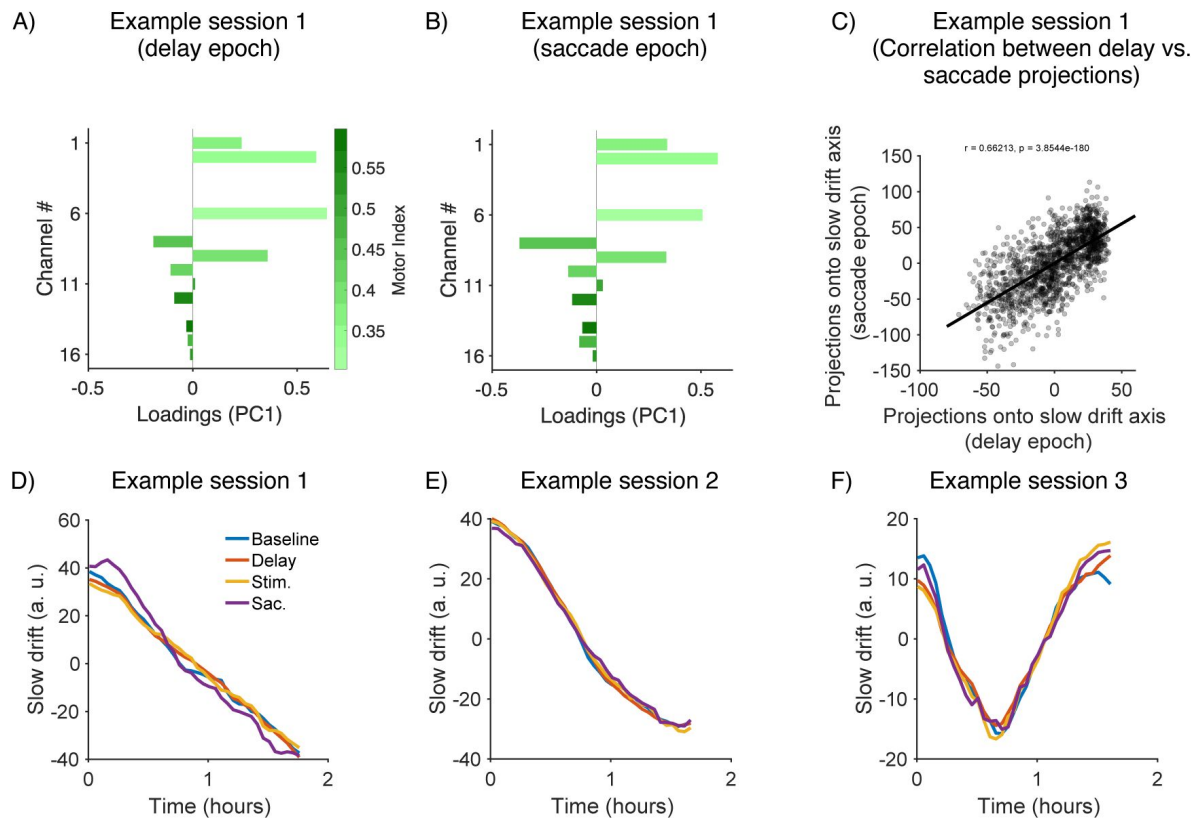


Figure 2 - figure supplement 1.

(A) Loadings for the SC slow drift axis for an example session when it was computed using spiking responses during the delay period (same session as **Figure 3A** in main manuscript). (B) Loadings for the SC slow drift axis from the same session when it was computed using spiking responses during the saccade epoch. (C) Scatter plot showing the correlation between projections onto the slow drift axis for the same session when the slow drift axis was computed using spiking responses during the delay and saccade epochs (one data point per trial). (D-F) Dynamics of slow drift for three example sessions when the SC slow drift axis was computed using spiking responses during the baseline, delay, visual and saccade epochs. Note that the session in (D) is the same as that shown in (A-C). Baseline = 100ms before the onset of the target stimulus; Delay = 600 to 1100ms after the offset of the target stimulus; Stim. = 25ms to 125ms after the onset of the target stimulus; Sac. = 25ms before to 25ms after the onset of the saccade.

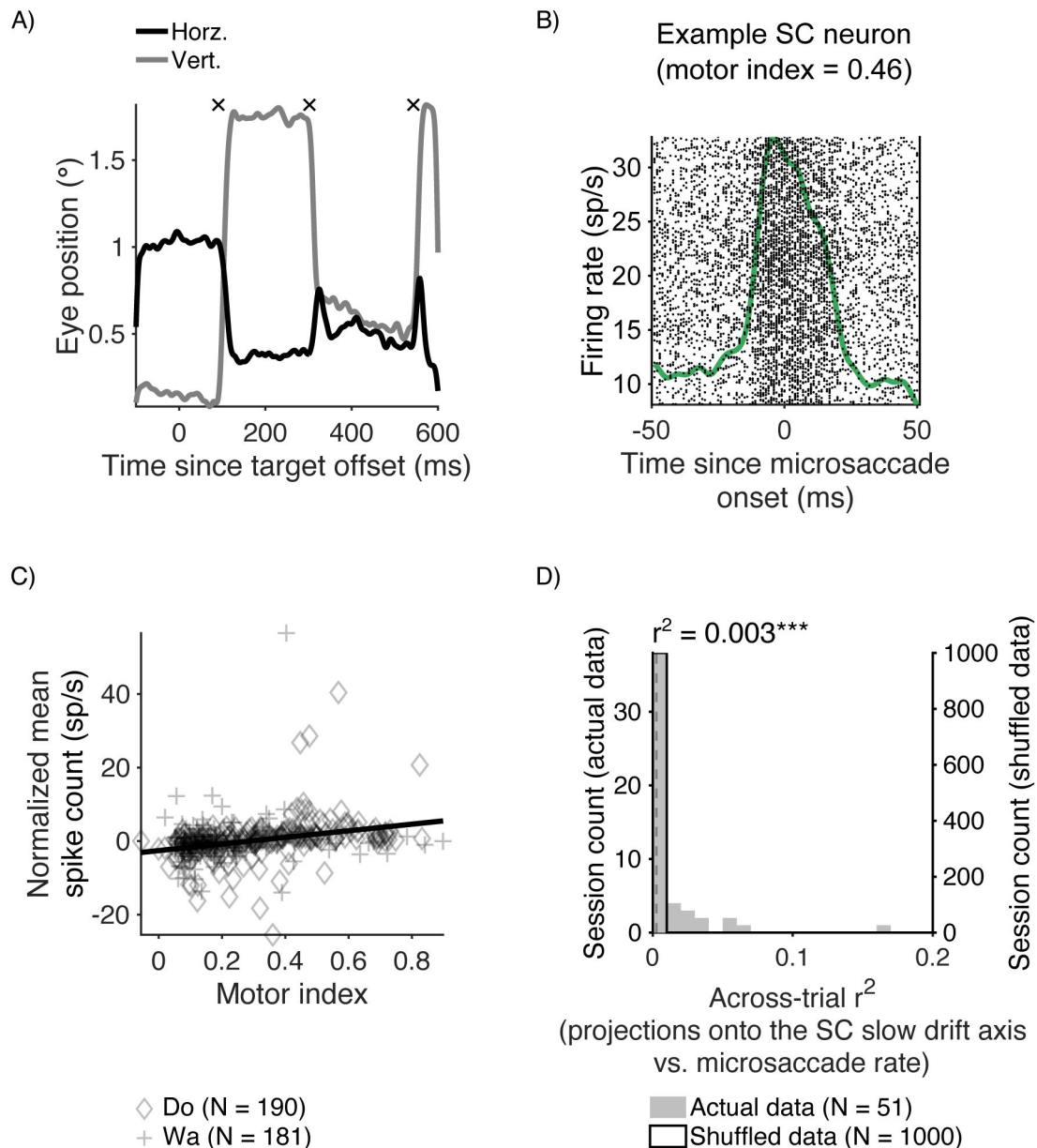


Figure 2 - figure supplement 2.

(A) Horizontal and vertical eye position on a single trial during the MGS task. Microsaccades were detected using the delay period using a velocity-based algorithm (Engbert and Kliegl, 2003) and are indicated by black crosses. (B) PSTHs aligned to microsaccade onset. Firing rates were computed in 1ms bins, averaged across trials and smoothed using a Gaussian function ($\sigma = 5\text{ms}$). (C) Scatter plot showing the relationship between normalized mean firing rate in and around the time of a microsaccade and motor index for each recorded SC neuron. For each neuron, mean firing rates were computed in the period spanning 25ms before and 25ms after a detected microsaccade, and were normalized by subtracting the mean activity across all trials during the same time period. This was done to control for the fact that some SC neurons (e.g. buildup neurons) have been found to increase their activity during the delay period on MGS tasks (Munoz and Wurtz, 1995). A multiple regression analysis was then performed to determine if normalized mean firing rate was associated with motor index. Results showed that the proportion of variance explained by the model, after controlling for possible intersubject differences, was 0.0791 ($F(2, 368) = 15.8089$; $p < 0.001$). Normalized mean firing rate was significantly associated with motor index such that it was higher for neurons that reside closer to the motor output ($t = 5.3747$; $p < 0.001$). (D) Histogram showing the proportion of variance explained between projections onto the SC slow drift axis and microsaccade rate. The median value across sessions is denoted by the dashed line. A p-value was computed by comparing the actual distribution of values to a null distribution (see Methods). $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

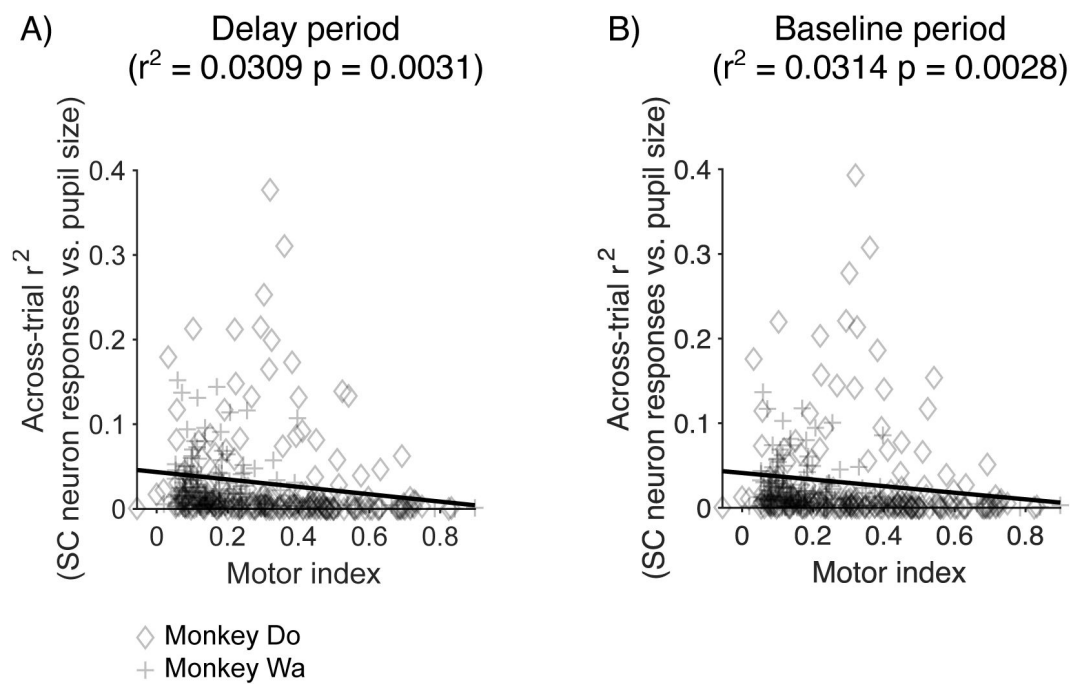


Figure 5 - figure supplement 1.

Scatter plot showing the strength of the relationship between individual SC neuron responses and pupil size.

In (A) pupil size was computed during the delay period whereas in (B) it was computed during the baseline period.

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Reviewer #1 (Public review):

Summary:

Johnston and Smith used linear electrode arrays to record from small populations of neurons in the superior colliculus (SC) of monkeys performing a memory-guided saccade (MGS) task. Dimensionality reduction (PCA) was used to reveal low-dimensional subspaces of population activity reflecting the slow drift of neuronal signals during the delay period across a recording session (similar to what they reported for parts of cortex: Cowley et al., 2020). This SC drift was correlated with a similar slow-drift subspace recorded from the prefrontal cortex, and both slow-drift subspaces tended to be associated with changes in arousal (pupil size). These relationships were driven primarily by neurons in superficial layers of the SC, where saccade sensitivity/selectivity is typically reduced. Accordingly, delay-period modulations of both spiking activity and pupil size were independent of saccade-related activity, which was most prevalent in deeper layers of the SC. The authors suggest that these findings provide evidence of a separation of arousal- and motor-related signals. The analysis techniques expand upon the group's previous work and provides useful insight into the power of large-scale neural recordings paired with dimensionality reduction. This is particularly important with the advent of recording technologies which allow for the measurement of spiking activity across hundreds of neurons simultaneously. Together, these results provide a useful framework for comparing how different populations encode signals related to cognition, arousal, and motor output in potentially different subspaces.

Comments on revised manuscript:

The authors have done a very good job of responding to all of the reviewers' concerns.

<https://doi.org/10.7554/eLife.99278.2.sa3>

Reviewer #2 (Public review):

Summary:

Neurons in motor-related areas have increasingly shown to carry also other, non-motoric signals. This creates a problem of avoidance of interference between the motor and non-motor-related signals. This is a significant problem that likely affects many brain areas. The specific example studied here is interference between saccade-related activity and slow-changing arousal signals in the superior colliculus. The authors identify neuronal activity related to saccades and arousal. Identifying saccade-related activity is straightforward, but arousal-related activity is harder to identify. The authors first identify a potential neuronal correlate of arousal using PCA to identifying a component in the population activity

corresponding to slow drift over the recording session. Next, they link this component to arousal by showing that the component is present across different brain areas (SC and PFC), and that it is correlated with pupil size, an external marker of arousal. Having identified an arousal-related component in SC, the authors show next that SC neurons with strong motor-related activity are less strongly affected by this arousal component (both SC and PFC). Lastly, they show that SC population activity pattern related to saccades and pupil size form orthogonal subspaces in the SC population.

Strengths:

A great strength of this research is the clear description of the problem, its relationship with the performed analysis and the interpretation of the results. the paper is very well written and easy to follow. An additional strength is the use of fairly sophisticated analysis using population activity.

Weaknesses:

(1) The greatest weakness in the present research is the fact that arousal is a functionally less important non-motoric variable. The authors themselves introduce the problem with a discussion of attention, which is without any doubt the most important cognitive process that needs to be functionally isolated from oculomotor processes. Given this introduction, one cannot help but wonder, why the authors did not design an experiment, in which spatial attention and oculomotor control are differentiated. Absent such an experiment, the authors should spend more time on explaining the importance of arousal and how it could interfere with oculomotor behavior.

(2) In this context, it is particularly puzzling that one actually would expect effects of arousal on oculomotor behavior. Specifically, saccade reaction time, accuracy, and speed could be influenced by arousal. The authors should include an analysis of such effects. They should also discuss the absence or presence of such effects and how they affect their other results.

(3) The authors use the analysis shown in Figure 6D to argue that across recording sessions the activity components capturing variance in pupil size and saccade tuning are uncorrelated. however, the distribution (green) seems to be non-uniform with a peak at very low and very high correlation specifically. The authors should test if such an interpretation is correct. If yes, where are the low and high correlations respectively? Are there potentially two functional areas in SC?

Comments on revised manuscript:

I remain somewhat concerned that the authors jump immediately into an analysis of the 'arousal-related' effects on SC activity. Before that, I would like to see a more detailed discussion justifying the use of pupil size alone (i.e., w/o other indicators such as RT) as indicative of fluctuations in general arousal that are causal to concomitant changes in SC activity. Instead, in its current form, the authors find changes in SC activity and describe them immediately as 'arousal-related'.

Other than this conceptual issue, I do not have major problems with the analysis per se.

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Reviewer #3 (Public review):

Summary:

This study looked at slow changes in neuronal activity (on the order of minutes to hours) in the superior colliculus (SC) and prefrontal cortex (PFC) of two monkeys. They found that SC

activity shows slow drift in neuronal activity like in the cortex. They then computed a motor index in SC neurons. By definition, this index is low if the neuron has stronger visual responses than motor response, and it is low if the neuron has weaker visual responses and stronger motor responses. The authors found that the slow drift in neuronal activity was more prevalent in the low motor index SC neurons and less prevalent in the high motor index neurons. In addition, the authors measured pupil diameter and found it to correlate with slow drifts in neuronal activity, but only in the neurons with lower motor index of the SC. They concluded that arousal signals affecting slow drifts in neuronal modulations are brain-wide. They also concluded that these signals are not present in the deepest SC layers, and they interpreted this to mean that this minimizes the impact of arousal on unwanted eye movements.

Strengths:

The paper is clear and well-written.

Showing slow drifts in the SC activity is important to demonstrate that cortical slow drifts could be brain-wide.

Weaknesses:

The authors find that the SC cells with the low motor index are modulated by pupil diameter. However, this could be completely independent of an "arousal signal". These cells have substantial visual sensitivity. If the pupil diameter changes, then their activity should be influenced since the monkey is watching a luminous display. So, in this regard, the fact that they do not see "an arousal signal" in the most motor neurons (through the pupil diameter analyses) is not evidence that the arousal signal is filtered out from the motor neurons. It could simply be that these neurons simply do not get affected by the pupil diameter because they do not have visual sensitivity. So, even with the pupil data, it is still a bit tricky for me to interpret that arousal signals are excluded from the "output layers" of the SC.

Of course, the general conclusion is that the motor neurons will not have the arousal signal. It's just the interpretation that is different in the sense that the lack of the arousal signal is due to a lack of visual sensitivity in the motor neurons.

I think that it is important to consider the alternative caveat of different amounts of light entering the system. Changes in light level caused by pupil diameter variations can be quite large. Please also note that I do not mean the luminance transient associated with the target onset. I mean the luminance of the gray display. It is a source of light. If the pupil diameter changes, then the amount of light entering to the visually sensitive neurons also changes.

Comments on revised manuscript:

The authors have addressed my first primary comment. For the light comment, I'm still not sure they addressed it. At the very least, they should explicitly state the possibility that the amount of light entering from the gray background can matter greatly, and it is not resolved by simply changing the analysis interval to the baseline pre-stimulus epoch. I provide more clear details below:

In line 194 of the redlined version of the article (in the Introduction), the citation to Baumann et al., PNAS, 2023 is missing near the citation of Jagadisan and Gandhi, 2022. Besides replicating Jagadisan and Gandhi, 2022, this other study actually showed that the subspaces for the visual and motor epochs are orthogonal to each other

Line 683 (and around) of the redlined version of the article (in the Results): I'm very confused here. When I mentioned visual modulation by changed pupil diameter, I did not mean the transient changes associated with the brief onset of the cue in the memory-guided saccade

task. I meant the gray background of the display itself. This is a strong source of light. If the pupil diameter changes across trials, then the amount of light entering the eye also changes from the gray background. Thus, visually-responsive neurons will have different amount of light driving them. This will also happen in the baseline interval containing only a fixation spot. The arguments made by the authors here do not address this point at all. So, please modify the text to explicitly state the possibility that the global luminance of the display (as filtered by the pupil diameter) alters the amount of light driving the visually-responsive neurons and could contribute to the higher effects seen in the more visual neurons.

The figures (everywhere, including the responses to reviewers) are very low resolution and all equations in methods are missing.

I'm very confused by Fig. 2 - supplement 2. Panel B shows a firing rate burst aligned to *microsaccade* onset. Does that mean you were in the foveal SC? i.e. how can neurons have a motor burst to the target of the memory-guided saccade and also for microsaccades? And which microsaccade directions caused such a burst? And what does it mean to compute the motor index and spike count for microsaccades in panel C? if you were in the proper SC location for the saccade target, then shouldn't you *not* get any microsaccade-related burst at all? This is very confusing to me and needs to be clarified

<https://doi.org/10.7554/eLife.99278.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

(1) The authors make fairly strong claims that "arousal-related fluctuations are isolated from neurons in the deep layers of the SC" (emphasis added). This conclusion is based on comparisons between a "slow drift axis", a low-dimensional representation of neuronal drift, and other measures of arousal (Figures 2C, 3) and motor output sensitivity (Figures 2B, 3B). However, the metrics used to compare the slow-drift axis and motor activity were computed during separate task epochs: the delay period (600-1100 ms) and a perisaccade epoch (25 ms before and after saccade initiation), respectively. As the authors reference, deep-layer SC neurons are typically active only around the time of a saccade. Therefore, it is not clear if the lack of arousal-related modulations reported for deep-layer SC neurons is because those neurons are truly insensitive to those modulations, or if the modulations were not apparent because they were assessed in an epoch in which the neurons were not active. A potentially more valuable comparison would be to calculate a slow-drift axis aligned to saccade onset.

The reviewer makes an important point that the calculation of an axis can depend critically on the time window of neuronal response. We find when considering this that the slow drift axis is less sensitive to this issue because it is calculated on time-averaged activity over multiple trials. In previous work we found that slow drift calculated on the stimulus evoked response in V4 was very well aligned to slow drift calculated on pre-stimulus spontaneous activity (Cowley et al, Neuron, 2020, Supplemental Figure 3A and 3B). To address this issue in the present data, we compared the axis computed for an example session for neural activity during the delay period and neural activity aligned to saccade onset. As shown new Figure 2 – figure supplement 1 in the revised manuscript, we found a similar lack of arousal-related modulations for deep-layer SC neurons when slow drift was computed using the saccade epoch (25ms before to 25ms after the onset of the saccade). Figure 2 – figure supplement 1A shows loadings for the SC slow drift axis when it was computed using spiking responses during the delay period (as in the main manuscript analysis). In contrast, Figure 2 – figure

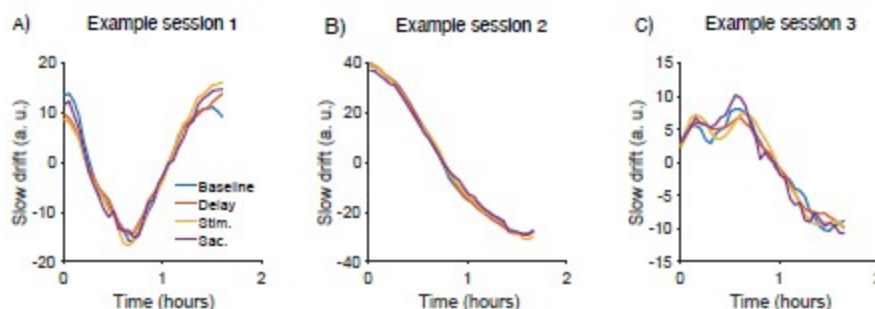
supplement 1B shows loadings from the same session when the SC slow drift axis was computed using spiking responses during the saccade epoch. The plots are highly similar and in both cases the loadings were weaker for neurons recorded from channels at the bottom of the probe which have a higher motor index. Finally, we found that projections onto the SC slow drift axis for this session were strongly correlated when the slow drift axis was computed using spiking responses during the delay period and the saccade epoch ($r = 0.66$, $p < 0.001$, Figure 1C). Taken together, these results suggest that arousal-related modulations are less evident in deep-layer SC neurons irrespective of whether slow drift was computed during the delay or saccade epoch (see also Public Reviews, Reviewer 1, Point 2).

(2) More generally, arousal-related signals may persist throughout multiple different epochs of the task. It would be worthwhile to determine whether similar "slow-drift" dynamics are observed for baseline, sensory-evoked, and saccade-related activity. Although it may not be possible to examine pupil responses during a saccade, there may be systematic relationships between baseline and evoked responses.

Similar to the point above, slow drift dynamics tend to be similar across different response epochs because they are averaged across many trials and seem to tap into responsivity trends that are robust across epochs. As shown in Author response image 1 below, and the Figure 2 – figure supplement 1 in the revised manuscript, similar dynamics were observed when the SC slow drift axis was computed using spiking responses during the baseline, delay, visual and saccade epochs. We did not investigate differences between baseline and evoked pupil responses in the current paper. However, these effects were characterized in one of our previous papers that focused exclusively on the relationship between slow drift and eye-related metrics (Johnston et al., 2022, Cereb. Cortex, Figure 6). In this previous work, we found a negative correlation between baseline and evoked pupil size. Both variables were significantly correlated with slow drift, the only difference being the sign of the correlation.

Author response image 1.

(A-C) Dynamics of slow drift for three example sessions when the SC slow drift axis was computed using spiking responses during the baseline, delay, visual and saccade epochs. Baseline = 100ms before the onset of the target stimulus; Delay = 600 to 1100ms after the offset of the target stimulus; Stim = 25ms to 125ms after the onset of the target stimulus; Sac = 25ms before to 25ms after the onset of the saccade.



Johnston R, Snyder AC, Khanna SB, Issar D, Smith MA (2022) The eyes reflect an internal cognitive state hidden in the population activity of cortical neurons. Cereb Cortex 32:3331–3346.

(3) The relationships between changes in SC activity and pupil size are quite small (Figures 2C & 5C). Although the distribution across sessions (Figure 2C) is greater than chance, they are nearly 1/4 of the size compared to the PFC-SC axis comparisons.

Likewise, the distribution of r^2 values relating pupil size and spiking activity directly (Figure 5) is quite low. We remain skeptical that these drifts are truly due to arousal and cannot be accounted for by other factors. For example, does the relationship persist if accounting for a very simple, monotonic (e.g., linear) drift in pupil size and overall firing rate over the course of an individual session?

Firstly, it is important to note that the strength of the relationship between projections onto the SC slow drift axis and pupil size ($r^2 = 0.06$) is within the range reported by Joshi et al. (2016, Neuron, Figure 3). They investigated the median variance explained between the spiking responses of individual SC neurons and pupil size and found it to be approximately 0.02 across sessions. Secondly, our statistical approach of testing the actual distribution of r^2 values against a shuffled distribution was specifically designed to rule out the possibility that the relationship between SC spiking responses and pupil size occurred due to linear drifts. The shuffled distribution in Figure 2C of the main manuscript represents the variance that can be explained by one session's slow drift correlated with another session's pupil, which would contain effects that occurred due to linear drifts alone. That the actual proportion of variance explained was significantly greater than this distribution suggests that the relationship between projections onto the SC slow drift axis and pupil size reflects changes in arousal rather than other factors related to linear drifts.

Joshi S, Li Y, Kalwani RM, Gold JI (2016) Relationships between Pupil Diameter and Neuronal Activity in the Locus Coeruleus, Colliculi, and Cingulate Cortex. Neuron 89:221–234.

(4) It is not clear how the final analysis (Figure 6) contributes to the authors' conclusions. The authors perform PCA on: (i) residual spiking responses during the delay period binned according to pupil size, and (ii) spiking responses in the saccade epoch binned according to target location (i.e., the saccade tuning curve). The corresponding PCs are the spike-pupil axis and the saccade tuning axis, respectively. Unsurprisingly, the spike-pupil axis that captures variance associated with arousal (and removes variance associated with saccade direction) was not correlated with a saccade-tuning axis that captures variance associated with saccade direction and omits arousal. Had these measures been related it would imply a unique association between a neuron's preferred saccade direction and pupil control- which seems unlikely. The separation of these axes thus seems trivial and does not provide evidence of a "mechanism...in the SC to prevent arousal-related signals interfering with the motor output." It remains unknown whether, for example, arousal-related signals may impact trial-by-trial changes in neuronal gain near the time of a saccade, or alter saccade dynamics such as acceleration, precision, and reaction time.

The reviewer makes a good point, and we agree that more evidence is needed to determine if the separation of the pupil size axis and saccade tuning axis is the mechanism through which cognitive and arousal-related signals can be intermixed in the SC. In the revised manuscript (lines 679-682), we have raised this as a possible explanation that necessitates further study rather than stating definitively that it is the exact mechanism through which these signals are kept separate. Our analysis here is similar to the one from Smoulder et al (2024, Neuron, Fig. 2F), in which the interactions between reward signals and target tuning in M1 were examined (and found to be orthogonal). While we agree with the reviewer that it may seem “trivial” for these axes to be orthogonal, it does not have to be so. If, for example, neural tuning curves shifted with changes in pupil size through gain changes that revealed tuning or affected tuning curve shape, there could be projections of the pupil axis onto the target tuning axis. Thus, while we agree with the reviewer that it appears sensible for these two axes to be orthogonal, our result is nonetheless a novel finding. We have edited the text in our revised manuscript, however, to make sure the nuance of this point is conveyed to the reader.

Smoulder AL, Marino PJ, Oby ER, Snyder SE, Miyata H, Pavlovsky NP, Bishop WE, Yu BM, Chase SM, Batista AP. A neural basis of choking under pressure. *Neuron*. 2024 Oct 23;112(20):3424-33.

Reviewer #2 (Public Review):

(1) The greatest weakness in the present research is the fact that arousal is a functionally less important non-motoric variable. The authors themselves introduce the problem with a discussion of attention, which is without any doubt the most important cognitive process that needs to be functionally isolated from oculomotor processes. Given this introduction, one cannot help but wonder, why the authors did not design an experiment, in which spatial attention and oculomotor control are differentiated. Absent such an experiment, the authors should spend more time explaining the importance of arousal and how it could interfere with oculomotor behavior.

Although attention does represent an important cognitive process, we did not design an experiment in which attention and oculomotor control are differentiated because attention does not appear to be related to slow drift. In our first paper that reported on this phenomenon, we investigated the effects of spatial attention on slow fluctuations in neural activity by cueing the monkeys to attend to a stimulus in the left or right visual field in a block-wise manner. Each block lasted ~20 minutes and we found that slow drift did not covary with the timing of cued blocks (see Figure 4A, Cowley et al., 2020, *Neuron*). Furthermore, there is a large body of work showing that arousal also impacts motor behavior leading to changes in a range of eye-related metrics (e.g., pupil size, microsaccade rate and saccadic reaction time - for review, see Di Stasi et al. 2013, *Neurosci. Biobehav. Rev.*). We also note that the terms attention and arousal are often used in nonspecific and overlapping ways in the literature, adding to some potential confusion here. Nonetheless, pupil-linked arousal is an important variable that impacts motor performance. This has now been stated clearly in the Introduction of the revised manuscript (lines 108-114) to address the reviewer's concerns and highlight the importance of studying how precise fixation and eye movements are maintained even in the presence of signals related to ongoing changes in brain state.

Cowley BR, Snyder AC, Acar K, Williamson RC, Yu BM, Smith MA (2020) Slow Drift of Neural Activity as a Signature of Impulsivity in Macaque Visual and Prefrontal Cortex. *Neuron* 108:551-567.e8.

(2) In this context, it is particularly puzzling that one actually would expect effects of arousal on oculomotor behavior. Specifically, saccade reaction time, accuracy, and speed could be influenced by arousal. The authors should include an analysis of such effects. They should also discuss the absence or presence of such effects and how they affect their other results.

As described above, several studies across species have demonstrated that arousal impacts motor behavior e.g., saccade reaction time, saccade velocity and microsaccade rate (for review, see Di Stasi et al. 2013, *Neurosci. Biobehav. Rev.*). This has been clarified in the Introduction of the revised manuscript to address the reviewer's concerns (lines 108-114). Our prior work (Johnston et al, *Cerebral Cortex*, 2022) shows that slow drift impacts several types of oculomotor behavior. Overall, these studies highlight the impact of arousal on eye movements as a robust effect, and support the present investigation into arousal and oculomotor control signals. While we agree reaction time, accuracy, and speed all can be influenced by arousal depending on task demands, the present study is focused on the connection between slow fluctuations in neural activity, linked to arousal, and different subpopulations of SC neurons.

Di Stasi LL, Catena A, Cañas JJ, Macknik SL, Martinez-Conde S (2013) Saccadic velocity as an arousal index in naturalistic tasks. *Neurosci Biobehav Rev* 37:968–975.

Johnston R, Snyder AC, Khanna SB, Issar D, Smith MA (2022) The eyes reflect an internal cognitive state hidden in the population activity of cortical neurons. *Cereb Cortex* 32:3331–3346.

(3) The authors use the analysis shown in Figure 6D to argue that across recording sessions the activity components capturing variance in pupil size and saccade tuning are uncorrelated. however, the distribution (green) seems to be non-uniform with a peak at very low and very high correlation specifically. The authors should test if such an interpretation is correct. If yes, where are the low and high correlations respectively? Are there potentially two functional areas in SC?

We agree with the reviewer that our actual data distribution was non-uniform. We examined individual sessions with high and low variance explained and did not find notable differences. One source of this variation has to do with session length. Longer sessions in principle should have a chance distribution of variance explained closer to zero because they contained more time bins. Given that we had no specific hypothesis for a non-uniform distribution, we have simply displayed the full distribution of values in our figure and the statistical result of a comparison to a shuffled distribution.

Reviewer #3 (Public Review):

(1) However, I am concerned about two main points: First, the authors repeatedly say that the "output" layers of the SC are the ones with the highest motor indices. This might not necessarily be accurate. For example, current thresholds for evoking saccades are lowest in the intermediate layers, and Mohler & Wurtz 1972 suggested that the output of the SC might be in the intermediate layers. Also, even if it were true that the high motor index neurons are the output, they are very few in the authors' data (this is also true in a lot of other labs, where it is less likely to see purely motor neurons in the SC). So, this makes one wonder if the electrode channels were simply too deep and already out of the SC? In other words, it seems important to show distributions of encountered neurons (regardless of the motor index) across depth, in order to better know how to interpret the tails of the distributions in the motor index histogram and in the other panels of Figure Supplement 1. I elaborate more on these points in the detailed comments below.

The reviewer makes a good point about the efferent signals from SC. It is true that electrical thresholds are often lowest in intermediate layers, though deep layers do project to the oculomotor nuclei (Sparks, 1986; Sparks & Hartwich-Young, 1989) and often intermediate and deep layers are considered to function together to control eye movements (Wurtz & Albano, 1980). As suggested by the reviewer, we have edited the text throughout the manuscript to say that slow drift was less evident in SC neurons with a higher motor index, as well as included the above references and points about the intermediate and deep layers (Lines 73–81). Aside from the question of which layers of the SC function as the “motor output”, the reviewer raises a separate and important question – are our deep recordings still in SC. Here, we can say definitively that they are. We removed neurons if they did not exhibit elevated (above baseline) firing rates during the visual or saccade epochs of the MGS task (see Methods section on “Exclusion criteria”). All included neurons possessed a visual, visuomotor or motor response, consistent with the response properties of neurons in the SC. In addition, we found a number of neurons well above the bottom of the probe with strong motor responses and minimal loadings onto the slow drift axis (see Figure 2 – figure supplement 1A), consistent with the reviewer’s comment that intermediate layer neurons are tuned for movement and play a role in saccade production.

Mohler CW, Wurtz RH. Organization of monkey superior colliculus: intermediate layer cells discharging before eye movements. *Journal of neurophysiology*. 1976 Jul 1;39(4):722-44.

Sparks DL. Translation of sensory signals into commands for control of saccadic eye movements: role of primate superior colliculus. *Physiol Rev*. 1986 Jan;66(1):118-71. doi: 10.1152/physrev.1986.66.1.118. PMID: 3511480.

Sparks DL, Hartwich-Young R. The deep layers of the superior colliculus. *Reviews of oculomotor research*. 1989 Jan 1;3:213-55.

Wurtz RH, Albano JE. Visual-motor function of the primate superior colliculus. *Annu Rev Neurosci*. 1980;3:189-226. doi: 10.1146/annurev.ne.03.030180.001201. PMID: 6774653.

(2) Second, the authors find that the SC cells with a low motor index are modulated by pupil diameter. However, this could be completely independent of an "arousal signal". These cells have substantial visual responses. If the pupil diameter changes, then their activity should be influenced since the monkey is watching a luminous display. So, in this regard, the fact that they do not see "an arousal signal" in most motor neurons (through the pupil diameter analyses) is not evidence that the arousal signal is filtered out from the motor neurons. It could simply be that these neurons simply do not get affected by the pupil diameter because they do not have visual sensitivity. So, even with the pupil data, it is still a bit tricky for me to interpret that arousal signals are excluded from the "output layers" of the SC.

The reviewer makes an important point about the SC's visual responses. Neurons with a low motor index are, conversely, likely to have a stronger visual response index. However, we do not believe that changes in luminance can explain why the correlation between SC spiking response and pupil size is weaker for neurons with a lower motor index. Firstly, the changes in pupil size observed in the current paper and our previous work are slow and occur on a timescale of minutes (Cowley et al., 2020, *Neuron*) and are correlated with eye movement measures such as reaction time and microsaccade rate (Johnston et al., 2022, *Cerebral Cortex*). This is in stark contrast to luminance-evoked changes in pupil size that occur on a timescale of less than a second. Secondly, as shown the new Figure 5 – figure supplement 1 in the revised manuscript, very similar results were found when SC spiking responses were correlated with pupil size during the baseline period, when only the fixation point was on the screen. Although the luminance of the small peripheral target stimulus can result in small luminance-evoked changes in pupil size, no changes in luminance occurred during the baseline period which was defined as 100ms before the onset of the target stimulus. In Figure 2 – figure supplement 1 and Author response image 1 above, we show that slow drift is the same whether calculated on the baseline response, delay period, or peri-saccadic epoch. Thus, the measurement of slow drift is insensitive to the precise timing of the selection of both the window for the spiking response and the window for the pupil measurement. If luminance were the explanation for the slow changes in firing observed in visually responsive SC neurons, it would require those neurons to exhibit robust, sustained tuned responses to the small changes in retinal illuminance induced by the relatively small fluctuations in pupil size we observed from minute to minute. We are aware of no reports of such behavior in visually-responsive neurons in SC. We have included these analyses and this reasoning in the revised manuscript on lines 478-495.

Reviewer#1 (Recommendations for the author):

(1) It would be useful to provide line numbers in subsequent manuscripts for reviewers.

Line numbers have been added in the revised version of the manuscript.

(2) Page #6; last sentence: "...even impact processing at the early to mid stages of the visuomotor transformation, without leading to unwanted changes in motor output." I do not believe the authors have provided evidence that arousal levels were not associated with changes in motor output.

As suggested by Reviewer 3 (see Public Reviews, Reviewer 3, Point 2), we have edited the text throughout the manuscript to say that slow drift was less evident in SC neurons with a higher motor index. This sentence in the revised manuscript now reads:

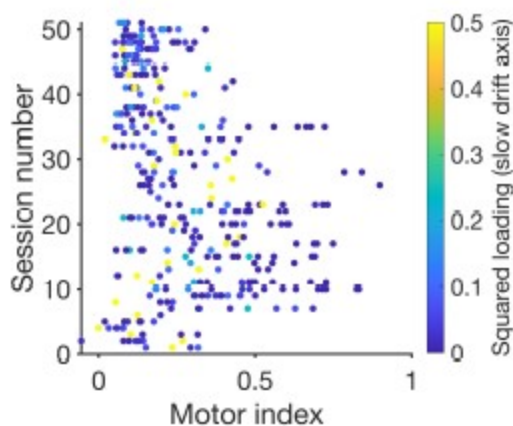
"This provides a potential mechanism through which signals related to cognition and arousal can exist in the SC, and even impact processing at the early to mid stages of the visuomotor transformation, without leading to unwanted changes in SC neurons that are linked to saccade execution."

(3) Page #8; last paragraph: Although deep-layer SC neurons may not have been obtained during every recording session, a summary of the motor index scores observed along the probe across sessions would be useful to confirm their assumptions.

See Author response image 2 below which shows the motor index of each recoded SC neuron on the x-axis and session number on the y-axis. The points are colored by to the squared factor loading which represents the variance explained between the response a neuron and the slow drift axis (see Figure 3B of the main manuscript). You can see from this plot that neurons with a stronger component loading (shown in teal to yellow) typically have a lower motor index whereas the opposite is true for neurons with a weaker component loading (shown in dark blue).

Author response image 2.

Scatter plot showing the motor index of each recorded neuron along with the session number in which it was recorded. The points are colored by to the squared factor loading for each neuron along the slow drift axis. Note that loadings above 0.5 (33 data points in total) have been thresholded at 0.5 so that we could effectively use the color range to show all of the slow drift axis loadings.



(4) Page #10; first paragraph: The authors should state the time window of the delay period used, since it may be distinct from the pupil analysis (first 200ms of delay).

This has been stated in the revised version of the manuscript. The sentence now reads:

“We first asked if arousal-related fluctuations are present in the SC. As in previous studies that recorded from neurons in the cortex (Cowley et al., 2020), we found that the mean spiking responses of individual SC neurons during the delay period (chosen at random on each trial from a uniform distribution spanning 600-1100ms, see Methods) fluctuated over the course of a session while the monkeys performed the MGS task (Figure 2A, left).”

(5) Page #10; second paragraph: Extra period at the end of a sentence: " most variance in the data..".

Fixed in the revised version of the manuscript.

(6) Page #12: "between projections onto the SC slow drift axis and mean pupil size during the first 200ms of the delay period when a task-related pupil response could be observed." What criteria was used to determine whether a task-related pupil response was observed?

This was chosen based on the results of a previous study in our lab that used the same memory-guided saccade task to investigate the relationship between slow drift and changes in based and evoked pupil size (see Johnston et al., 2022, Cereb. Cortex, Figure 6B). The period was chosen based on plotting the average pupil size aligned on different trial epochs. As we show in Figure 5-figure supplement 3 above, the pupil interactions with slow drift did not depend on the particular time window of the pupil we chose.

(7) Page #14; Figure 2A: The axes for the individual channels are strangely floating and quite different from all other figures. Please label the channel in the figure legend that was used as an example of the projected values onto the slow drift axis.

The figure has been changed in the revised version of the manuscript so that the tick mark denoting zero residual spikes per second is on the top layer of each plot. A scale bar was chosen instead of individual axes to reduce clutter in the figure as it was used to demonstrate how slow drift was computed. Residual spiking responses from all neurons were projected on the slow drift axis to generate the scatter plot in the bottom right-hand corner of Figure 2A. There is no single neuron to label.

(8) Page #16: "These results demonstrate that even though arousal-related fluctuations are present in the SC, they are isolated from deep-layer neurons that elicit a strong saccadic response and presumably reside closer to the motor output." In line with our major comments, lack of arousal-related activity during the delay period is meaningless for deep-layer SC neurons that are generally inactive during this time. It does not imply that there is no arousal signal!

Addressed in Public Reviews, Reviewer 1, Point 1 & 2. We found a similar lack of arousal-related modulations reported for deep-layer SC neurons when slow drift was computed using the saccade epoch (Figure 1 above). In addition, similar dynamics were observed when the SC slow drift axis was computed using spiking responses during the baseline, delay, visual and saccade period (Figure 2).

(9) Page #18: "These findings provide additional support for the hypothesis that arousalrelated fluctuations are isolated from neurons in the deep layers of the SC." The same criticism from above applies.

Addressed in Public Reviews, Reviewer 1, Point 1 & 2.

(10) Page #20; paragraph 3: "Taken together, the findings outlined above..." Would be useful to be more specific when referring to "activity"; e.g., "...these neurons did not exhibit large fluctuations in delay-period activity over time".

This sentence has been changed in the revised manuscript in light of the reviewer's comments. It now reads:

"In addition to being more weakly correlated with pupil size, the spiking responses of these neurons did not exhibit large fluctuations over time (Figure 2), and when considering the neuronal population as a whole, explained less variance in the slow drift axis when it was computed using population activity in the SC (Figure 3) and PFC (Figure 4)."

Reviewer #3 (Recommendations for the author):

The paper is clear and well-written. However, I am concerned about two main points:

(1) First, the authors repeatedly say that the "output" layers of the SC are the ones with the highest motor indices. This might not necessarily be accurate. For example, current thresholds for evoking saccades are lowest in the intermediate layers, and Mohler & Wurtz 1972 suggested that the output of the SC might be in the intermediate layers. Also, even if it were true that the high motor index neurons are the output, they are very few in the authors' data (this is also true in a lot of other labs, where it is less likely to see purely motor neurons in the SC). So, this makes one wonder if the electrode channels were simply too deep and already out of the SC. In other words, it seems important to show distributions of encountered neurons (regardless of motor index) across depth, in order to better know how to interpret the tails of the distributions in the motor index histogram and in the other panels of the figure supplement 1. I elaborate more on these points in the detailed comments below.

Addressed in Public Reviews, Reviewer 3, Point 1.

(2) Second, the authors find that the SC cells with a low motor index are modulated by pupil diameter. However, this could be completely independent of an "arousal signal". These cells have substantial visual responses. If the pupil diameter changes, then their activity should be influenced since the monkey is watching a luminous display. So, in this regard, the fact that they do not see "an arousal signal" in most motor neurons (through the pupil diameter analyses) is not evidence that the arousal signal is filtered out from the motor neurons. It could simply be that these neurons simply do not get affected by the pupil diameter because they do not have visual sensitivity. So, even with the pupil data, it is still a bit tricky for me to interpret that arousal signals are excluded from the "output layers" of the SC.

Addressed in Public Reviews, Reviewer 3, Point 2.

(3) I think that a remedy to the first point above is to change the text to make it a bit more descriptive and less interpretive. For example, just say that the slow drifts were less evident among the neurons with high motor index.

We thank the reviewer for this suggestion (see Public Reviews, Reviewer 3, Point 1).

(4) For the second point, I think that it is important to consider the alternative caveat of different amounts of light entering the system. Changes in light level caused by pupil diameter variations can be quite large.

We thank the reviewer for this suggestion (see Public Reviews, Reviewer 3, Point 2).

(5) Line 31: I'm a bit underwhelmed by this kind of statement. i.e. we already know that cognitive processes and brain states do alter eye movements, so why is it "critical" that high precision fixation and eye movements are maintained? And, isn't the next sentence already nulling this idea of criticality because it does show that the brain state alters the SC neurons? In fact, cognitive processes are already known to be most prevalent in the intermediate and deep layers of the SC.

It seems clear that while cognitive state does affect eye movements, it is desirable to have some separation between cognitive state and eye movement control. Covert attention, for instance, is precisely a situation where eye movement control is maintained to avoid overt saccades to the attended stimulus, and yet there are clear indications of attention's impact on microsaccades and fixation. We stand by our statement that an important goal of vision is to have precise fixation and movements of the eye, and yet at the same time the eyes are subject to numerous influences by cognitive state.

(6) Line 65: it is better to clarify that these are "functional layers" because there are actually more anatomical layers.

We have edited this sentence in the revised version of the manuscript so that it now reads:

“The role of these projections in the visuomotor transformation depends on the functional layer of the SC in which they terminate”.

(7) Line 73: this makes it sound like only the deepest layers are topographically organized, which is not true. Also, as early as Mohler & Wurtz, 1972, it was suggested that the intermediate layers have the biggest impacts downstream of the SC. This is also consistent with electrical microstimulation current thresholds for evoking saccades from the SC.

We have addressed the reviewers' comments about the intermediate layers having the biggest impact downstream of the SC in Public Reviews, Reviewer 3, Point 1. Furthermore, line 73 has been changed in the revised manuscript so that it now reads:

“As is the case for neurons in the superficial and intermediate layers, they [SC motor neurons] form a topographically organized map of visual space (White et al. 2017; Robinson 1972; Katnani and Gandhi 2011)”.

(8) Line 100: there is an analogous literature regarding the question of why unwanted muscle contractions do not happen. Specifically, in the context of why SC visual bursts do not automatically cause saccades (which is a similar problem to the ones you mention about cognitive signals interfering by generating unwanted eye movements), both Jagadisan & Gandhi, Curr Bio, 2022 and Baumann et al, PNAS, 2023 also showed that SC population activity not only has different temporal structure (Jagadisan & Gandhi) but also occupy different subspaces (Baumann et al) under these two different conditions (visual burst versus saccade burst). This is conceptually similar to the idea that you are mentioning here with respect to arousal. So, it is worth it to mention these studies here and again in the discussion.

We are grateful to the reviewer for these suggestions and have included text in the Introduction (Lines 125-128) and Discussion (Lines 678-682) of the revised manuscript along with the references cited above.

(9) Line 147: as mentioned above, it is now generally accepted that there are quite a few "pure" motor neurons in the SC. This is consistent with what you find. E.g. Baumann et al., 2023. And, again see Mohler and Wurtz in the 1970's. So, I wonder how useful it is to go too much into this idea of the deeper motor neurons (e.g. the correlations in the other panels of the Figure 1 supplement).

This is related to the reviewer's comment that the output of the SC might be in the intermediate layers. This concern has been addressed in Public Reviews, Reviewer 3, Point 1.

(10) Figure 1 should say where the RF was for the shown spike rasters. i.e. were these the same saccade target across trials? And where was that location relative to the RF? It would help also in the text to say whether the saccade was always to the RF center or whether you were randomizing the target location.

We centered the array of saccade targets using the microstimulation-evoked eye movement for SC (see Methods section "Memory-guided saccade task") to find the evoked eccentricity, and then used saccade targets with equal spacing of 45 degrees starting at zero (rightward saccade target). We did not do extensive RF mapping beyond this microstimulation centering. In Figure 1, the spike rasters are shown for a target that was visually identified to be within the neuron's RF based on assessing responses to all 8 target angles. We have added information about this to the figure caption.

(11) Line 218: but were there changes in the eye movement statistics? For example, the slow drift eye movements during fixation? Or even the microsaccades?

Addressed in Public Reviews, Reviewer 2, Point 2.

(12) Line 248: shuffling what exactly? I think that more explanation would be needed here.

Addressed in Public Reviews, Reviewer 1, Point 3.

(13) Line 263: but isn't this reflecting a sensory transient in the pupil diameter, since the target just disappeared?

Addressed in Public Reviews, Reviewer 3, Point 2.

(14) Line 271: I suspect that slow drift eye movements (in between microsaccades) would show higher correlations. Not sure how well you can analyze those with a video-based eye tracker.

We agree that fixational drift would be a worthwhile metric, but it is not one we have focused on here and to our knowledge does require higher precision tracking.

(15) Line 286: again, see above about similar demonstrations with respect to the visual and motor burst intervals, which clearly cause the same problem (even stronger) as the one studied here.

See reply, including Figure 2.

(16) Line 330: again, I'm not sure deeper necessarily automatically means closer to the output. For example, current thresholds for evoked saccades grow higher as you go deeper. Maybe the authors can ask their colleague Neeraj Gandhi about this point

specifically, just to be safe. Maybe the safest would be to remain descriptive about the data, and just say something like: arousal-related fluctuations were absent in our deepest recorded sites.

Addressed in Public Reviews, Reviewer 3, Point 1.

(17) Line 332: likewise, statements like this one here would be qualified if the output was the intermediate layers.....anyway if I understand what I read so far in the paper, the signal will be anyway orthogonal to the motor burst population subspace. So, maybe there's no need to emphasize that it goes away in the very deepest layers.

See reply above, Public Reviews, Reviewer 1, Point 4.

(18) Figure 3A: related to the above, I think one issue could be that the deeper contacts might already be out of the SC. Maybe some cell count distribution from each channel should help in this regard. i.e. were you finding way fewer saccade-related neurons in the deepest channels (even though the few that you found were with high motor index)? If so, then wouldn't this just mean that the channel was too deep? I think there needs to be an analysis like this, to convince readers that the channels were still in the SC. Ideally, electrical stimulation current thresholds for evoking saccades at different depths would be tested, but I understand that this can be difficult at this stage.

Addressed in Public Reviews, Reviewer 3, Point 1.

(19) I keep repeating this because in general, cognitive effects are stronger in the intermediate/deeper layers than in the superficial layers. If these interfere with eye movements like arousal, then why should arousal be different?

Few studies have investigated the effects of attention on “pure” movement SC neurons that only discharge during a saccade. One study, which we cited in Introduction (Ignashchenkova et al., 2004, Nat. Neurosci.), found significant differences in spiking responses between trials with and without attentional cueing for visual and visuomotor neurons. No significant difference was found for motor neurons, consistent with our hypothesis that signals related to cognition and arousal are kept separate from saccade-related signals in the SC.

(20) The problem with Figure 5 and its related text is that the neurons with low motor index are additionally visual. So, of course, they can be modulated if the pupil diameter changes!

Addressed in Public Reviews, Reviewer 3, Point 2.

(21) I had a hard time understanding Figure 6.

See reply above, Public Reviews, Reviewer 1, Point 4.

(22) Line 586: these cells have more visual responses and will be affected by the amount of light entering the eye.

Addressed in Public Reviews, Reviewer 3, Point 2.

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